Shared Gradient Clipping Alters the Effects of Reward Scaling on Finite-Budget Coordination-Collapse Crossings in PPO*

Anonymous authors
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Abstract

Cooperative multi-agent learners can collapse into low-cooperation behavior when local incentives outweigh the team objective, and reward design is a common remedy. We measure where collapse sets in along a local/team reward-weighting dial by an operational, finite-budget crossing w_{50}, the weight at which a fitted collapse probability passes 50%, estimated from behavioral labels with training-seed-block intervals. A symmetric rescaling of the reward should not change what is learned, yet in default Stable-Baselines3 PPO on an inventory network, scaling both reward branches by seven moves the crossing by +0.357 [+0.249, +0.464]. The cause is an implementation detail: actor and critic share one global gradient-norm clip, the unnormalized critic's gradient dominates it, and the actor's applied gradient and parameter update shrink to 0.12 and 0.39 of baseline. Clipping actor and critic separately restores the update and reduces the shift by −0.207 [−0.336, −0.085]; reward normalization leaves no detectable symmetric shift (−0.049 [−0.117, +0.017]), and normalizing only the critic's targets shrinks the remainder by an amount ten seeds cannot resolve. With the coupling removed, a loss-only multiplier still moves the crossing relative to the symmetric control (+0.237 [+0.136, +0.346]), through 3.6M transitions. This asymmetric finding is conditional: against a symmetric control matched to its reward scale but not its mean, it is separable only when the critic's targets are normalized (+0.268 [+0.211, +0.317]), not with raw targets (+0.050 [+0.000, +0.100]). Results come from two small environments and one PPO implementation; w_{50} is a comparative instrument, not a critical point.

1 Introduction

A team of learning agents can fail even when the task has a good cooperative solution: instead of learning complementary behavior, agents can settle into locally safe, low-cooperation behavior. We call this *coordination collapse*; in the MARL literature its endpoint is the *lazy-agent* policy (Liu et al., 2023; Shi et al., 2023), and the underlying private-versus-collective conflict is the one studied in sequential social dilemmas (Leibo et al., 2017). A common design response is to give each learner part of the team's reward. We study that choice through a local/team *reward-weighting* dial, $x_i(w) = (1-w)\, r_i^{\text{local}} + w\, r^{\text{global}}$, and summarize where collapse sets in by a finite-budget behavioral crossing w_{50}: the weight at which a fitted collapse probability passes 50%, estimated from pre-specified evaluation labels with seed-paired uncertainty.

Reward-transform comparisons along this dial look as if they should be clean. Multiplying every reward by a positive constant leaves every optimal policy unchanged, and with per-minibatch advantage normalization it leaves the policy-gradient step of PPO (Schulman et al., 2017) approximately unchanged as well. A symmetric rescaling should therefore not move the crossing, whereas a transform that weights losses more than gains changes which outcomes the learner prefers and could move it. Practical actor–critic implementations break this separation. The critic's regression targets are usually not normalized, so their gradient grows with the

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reward; and Stable-Baselines3 PPO (Raffin et al., 2021) by default clips the norm of the *joint* actor–critic gradient. When the critic's gradient dominates that joint norm, the clip shrinks the actor's step in proportion to the reward scale. A change in reward magnitude can then change what is learned even though it does not change what is optimal.

It is known that PPO's results depend on implementation details (Engstrom et al., 2020; Andrychowicz et al., 2021; Henderson et al., 2018), that value-target scale can be normalized away (van Hasselt et al., 2016), and that cooperative PPO implementations normalize value targets (Yu et al., 2022). What has not been shown is that one such detail, a global clip shared by an actor and an unnormalized critic, changes the *conclusion* of a reward-design comparison in MARL, nor how large that change is on a behavioral threshold measured with paired uncertainty. We ask three questions. **RQ1:** Does a symmetric reward rescaling move the finite-budget crossing under default PPO, and through which optimizer quantity: pre-clip gradient, applied gradient, or parameter update? **RQ2:** Which interventions remove that shift, and what remains when they are applied? **RQ3:** With the pathway removed, does a loss-only multiplier still move the crossing relative to symmetric controls, including one matched to its reward scale?

The contributions, each tied to the evidence that supports it, are: (i) *a mechanism for an optimizer confound* (Sections 4 and 6; Table 4). Under default SB3 PPO, symmetric $\times 7$ scaling moves the crossing by $+0.357$ $[+0.249, +0.464]$. Per-minibatch logging shows the critic taking the shared clip: the actor's applied gradient under symmetric scaling is 0.12 of the baseline's and its parameter update 0.39. Clipping actor and critic separately restores the update (0.95) and reduces the shift by -0.207 $[-0.336, -0.085]$. The remainder, which clipping does not touch, is not detectable under reward normalization; normalizing only the critic's targets shrinks it by an amount ten seeds cannot separate from zero. (ii) *a measurement protocol with checked uncertainty* (Sections 5 and 9): Firth logistic crossings with training-seed-block intervals, pre-registered experiments (4,060 trajectories, of which 2,720 address the optimizer mechanism), and a simulation showing that the nominal 95% intervals cover at 0.88–0.92; every principal verdict is unchanged at a coverage-calibrated 99% level. (iii) *a conditional asymmetric effect* (Section 7). With the coupling removed and rewards normalized, a loss-only multiplier raises the crossing relative to the symmetric control ($+0.237$ $[+0.136, +0.346]$), through 3.6M transitions. Against a symmetric control matched to its reward scale, the effect is separable only when the critic's targets are normalized; with raw rewards and a raw-target critic it may be an effect of the reward's moments. In Level-Based Foraging an asymmetric transform also moves the crossing, through a different incentive route (Section 8). We use "boundary" and "crossing" only in this operational, finite-budget sense, not as claims of phase-like criticality.

Section 2 positions the work; Section 3 defines the reward, transforms, label and estimand; Section 4 derives the shared-clip pathway; Section 5 sets out the experiments and statistics; Sections 6 and 7 report symmetric and asymmetric transforms; Section 8 turns to a second environment; Section 9 collects robustness checks; Sections 10 and 11 state what the results do and do not establish.

2 Related Work

Cooperative MARL, studied since independent Q-learning on shared tasks (Tan, 1993; Buşoniu et al., 2008), inherits the lazy-agent and relative-overgeneralization failures in which agents abandon cooperation for a degenerate policy (Liu et al., 2023; Shi et al., 2023; Tampuu et al., 2017). The standard remedy is centralized credit assignment under CTDE, a centralized critic or factorized value function (Lowe et al., 2017; Yu et al., 2022; Sunehag et al., 2018; Rashid et al., 2018; Foerster et al., 2018; Son et al., 2019); related work ties the failure to decentralized exploration and parameter sharing, and locates an empirical collapse threshold over payoff harshness for each exploration level, averaged over 30 seeds (Weng & Lee, 2026), or shows decentralized shaping cannot match centralized baselines (Akella, 2025). This establishes *that* centralization helps, *how* to implement it, and that collapse thresholds can be measured. It does not provide a 50% collapse-probability crossing along a local/team reward-weighting dial with seed-paired intervals, or test how reward transforms and optimizer implementation move it. Combined individual-and-team reward design has direct precedent: DE-MADDPG separates a per-agent and a team critic for such mixtures (Sheikh & Bölöni, 2020), packet-routing work studied the lazy-versus-selfish trade-off with mixed and adaptive local/global signals (Mao et al., 2020), IRAT learns individual- and team-reward policies with mutual discrepancy constraints (Wang et al.,

2022), and CLOT balances individual and team optimization through a Lagrangian multiplier (Zhang et al., 2026). These set or adapt the mixture as a design choice rather than *locating* a crossing with uncertainty.

Statistically, a 50% crossing is an effective-dose or psychometric-threshold estimand. Dose–response analysis routinely estimates ED50 values and differences between them (Ritz et al., 2015), and psychometric work studies bootstrap uncertainty and grid placement (Wichmann & Hill, 2001). We import that logic into trained multi-agent populations; our contribution lies in the experimental unit, collapse label, seed-block resampling, bracketing diagnostics and paired comparison of reward designs. Treating collective learning as a dynamical system with critical points has precedent in phase diagrams over environmental axes (Yamaguchi, 2025), metastable opinion-dynamics boundaries (Capannoli et al., 2026), and mean-field limits of multi-agent learning (Hu et al., 2019; Barfuss et al., 2019; Leonte et al., 2025); ours is a reward-structure threshold estimated from trained populations rather than asserted to be a dynamical critical point.

PPO's behavior depends on implementation details that are not part of the algorithm's definition (Engstrom et al., 2020; Andrychowicz et al., 2021; Henderson et al., 2018), among them reward or value normalization (van Hasselt et al., 2016) and gradient-norm clipping, which was introduced to bound exploding gradients (Pascanu et al., 2013); cooperative PPO implementations normalize value targets by default (Yu et al., 2022). Andrychowicz et al. (2021) vary, among many choices, whether actor and critic share a network and how gradients are clipped, and measure returns. We hold the algorithm fixed and ask a narrower question: whether one combination of these defaults, a global clip shared by actor and critic without value normalization, changes the answer to a reward-design question, and through which of the gradient, applied-gradient and parameter-update quantities it does so.

Human preference structure has entered collective learning only narrowly. Hughes et al. (2018) show inequity aversion improves cooperation within the cooperative-AI agenda (Dafoe et al., 2020; Hammond et al., 2025; Barfuss et al., 2020); KTO aligns language models with a Kahneman–Tversky value function carrying an explicit loss-aversion coefficient (Ethayarajh et al., 2024); risk-sensitive RL optimizes tail outcomes (Qiu et al., 2021; Shen et al., 2023; Prashanth, 2014; Tamar et al., 2015); and prospect theory has entered single-agent RL with known calibration concerns (Prashanth et al., 2016; Rabin, 2000; Tversky & Kahneman, 1992). Mengesha et al. (2025) find loss aversion non-monotone in an evolutionary quantal-response model, and Lalmohammed et al. (2025) bring CPT into cooperative MARL inside the actor–critic update, varying value and probability-weighting components jointly with the optimizer. Unlike inequity aversion, which penalizes the *difference* between own and peer payoffs, our transform acts on the already-blended scalar against a fixed reference, so it changes the magnitude and sign-weighting of the training signal rather than the local-versus-collective trade-off itself.

3 Problem Formulation and the Finite-Budget Crossing

We study *reward-mixed* cooperative MARL, in which each learner's scalar reward is the blend $x_i(w) = (1-w)\, r_i^{\text{local}} + w\, r^{\text{global}}$, $w \in [0,1]$. On the supply chain the team reward sums the five trained nodes' profits, so $x_i(w) = r_i^{\text{local}} + w \sum_{j \neq i} r_j^{\text{local}}$: node i's own profit keeps unit weight and w adds teammates' profits. We do not rescale r_i^{local} or r^{global} at each w, so a given w means the same mixture in every condition; the scale of x_i therefore varies with w, which matters under the optimizer coupling studied in Section 6. Each experiment varies one factor at a time. The primary substrate is `NetworkManagement-v1` (Hubbs et al., 2020), a six-node multi-echelon inventory network (Oroojlooyjadid et al., 2022; Gijsbrechts et al., 2022; Leluc et al., 2023) wrapped with PettingZoo/SuperSuit (Terry et al., 2021; 2020); the second environment is Level-Based Foraging (Section 8).

The boundary estimate follows four steps: fix the environment, learner, horizon, and collapse label; train matched policies across a grid of w; label each run from pre-specified *evaluation* behavior, not training reward; and fit a monotone binary model $P(\text{collapse} \mid w)$, reporting the midpoint w_{50} with bootstrap confidence intervals.

Information structure. The supply-chain experiments train one policy, shared by the five trainable nodes, on each node's own blended reward. Every node's policy input is the *same* full observation of the

network state; there is no agent identifier, and actions are padded to the largest per-node action dimension, each node using its leading entries. Under deterministic evaluation the five nodes therefore receive the same action vector. The setting is thus a fully observed, homogeneous shared controller trained on per-node mixed rewards, not a decentralized partially observed team: collapse here is the shared policy settling on the lazy action, and nothing below is evidence about decentralized information. In Level-Based Foraging each agent's actor sees only its own observation and a centralized critic sees the global state (Section 8).

Reward transforms. With zero as the gain/loss reference, $v(x) = x^\alpha$ for $x \geq 0$ and $-\lambda(-x)^\beta$ for $x < 0$. The exponents $\alpha, \beta \in (0, 1]$ control *curvature*; the multiplier $\lambda > 1$ scales the loss branch. For $a > 0$, $|v(-a)|/v(a) = \lambda a^{\beta-\alpha}$, which equals λ at every magnitude only when $\alpha = \beta$, as in all our experiments. The transform is applied after blending,

$$r_i^{\mathrm{PT}} = v\big((1-w)\, r_i^{\mathrm{local}} + w\, r^{\mathrm{global}}\big)\,. \tag{1}$$

Controls use $r_i^{\mathrm{PT}} = v((x_i - z_0)/s_{\mathrm{in}})/s_{\mathrm{out}}$ with no addition of z_0 after transformation; Equation 1 has $z_0 = 0$ and $s_{\mathrm{in}} = s_{\mathrm{out}} = 1$. The scale controls of Section 6 use the frozen empirical-mean reference $z_0 = -2.6281398333$ (Appendix A) with $u = x - z_0$ and $\phi(u) = \mathrm{sign}(u)|u|^{.88}$: arm B= $\phi(u)$ (curvature only), A= $v_7(u)$ (losses $\times 7$), S= $7\phi(u)$ (both branches $\times 7$, i.e. $s_{\mathrm{out}} = 1/7$), and R= $3.451\,\phi(u)$ (both branches scaled by A's pooled RMS ratio). We omit the probability-weighting channel of full CPT. Unlike Lalmohammed et al. (2025), the transform sits outside the optimizer, so within a fixed optimizer configuration matched runs differ only in the scalar training signal.

Collapse label. A *collapse label* is a pre-specified binary evaluation rule scored on held-out evaluation episodes. In the supply chain, collapse is $\mathbf{1}[\bar{S} < 10 \wedge \bar{\pi} \leq -127.1]$, where $\bar{\pi}$ is the total profit of nodes 2–6 averaged over ten deterministic evaluation episodes (seeds $10s, \ldots, 10s + 9$ for training seed s) and $\bar{S}$ is the mean over the five trained nodes of the deterministic base-stock target $250(a_1 + 1)$ chosen at the reset of the first evaluation episode, where a_1 is the node's first action component. Node 1 is scripted to order 20 units, split equally across its action entries. Collapsed runs sit at the lazy-policy profit floor (≈ -128), so the profit clause is a redundancy check on $\bar{S} < 10$; threshold variants of $\pm 20\%$ change w_{50} by at most 0.015. Across all 1,280 E1 runs only 1 has $\bar{S}$ within one unit of the threshold, so the label is not a fine cut through a continuum (Figure 13, Appendix E).

Estimand. We write the estimand as $w_{50}(T; E, A, V, U, L, Q)$: training budget, environment, learner (including its optimizer configuration), transform, reward units, collapse label and evaluation protocol. It is a fitted 50% crossing, not a safe operating point, and a rightward shift means more team weight is needed to reach it. We fit $P(\text{collapse} \mid w) = \sigma(b_0 + b_1 w)$ and report $w_{50} = -b_0/b_1$ (Section 5). Collapse becomes less likely as w grows, so $b_1 < 0$ in every fit we report.

4 The Shared-Clip Pathway in PPO

Learner and optimizer. The supply-chain learner is parameter-shared PPO in Stable-Baselines3 2.8.0 (Raffin et al., 2021): one policy across nodes, separate actor and critic MLPs, a single Adam optimizer ($\epsilon = 10^{-5}$, SB3's implicit default) over both, advantages normalized per minibatch, no value normalization, and one global gradient-norm clip of 0.5 applied to the concatenated actor and critic gradient. Because the actor's advantages are normalized but the critic's targets are not, a larger reward scale enlarges only the value-loss gradient, which then dominates the shared clip. Section 6 varies exactly this coupling. Training budgets count requested agent transitions; with five trained lanes and 128-step rollouts each rollout adds 640 transitions, so 300k requested transitions execute 300,160 (Appendix A).

How reward scale reaches the actor. Write the PPO loss as $\mathcal{L} = \mathcal{L}_\pi(\theta_a) + c_v\, \mathcal{L}_V(\theta_c)$ with value-loss coefficient $c_v = 0.5$ and no entropy term, where the actor parameters θ_a and critic parameters θ_c are disjoint (the networks share no trainable parameter). On a minibatch let $g_a = \nabla_{\theta_a} \mathcal{L}$ and $g_c = \nabla_{\theta_c} \mathcal{L}$. SB3 rescales

the concatenated gradient to norm at most $\tau = 0.5$,

$$\tilde{g} = \kappa\,(g_a, g_c), \qquad \kappa = \min\Big(1,\ \frac{\tau}{\sqrt{\|g_a\|^2 + \|g_c\|^2}}\Big), \qquad \text{so} \qquad \|\tilde{g}_a\| = \kappa\|g_a\| \approx \tau\,\frac{\|g_a\|}{\|g_c\|} \ \text{ when } \|g_c\| \gg \|g_a\|. \tag{2}$$

Multiply every reward by $s > 0$. Returns, value targets and, at a fixed relative fit, the value error scale by s, so $\|g_c\| = c_v\|\nabla_{\theta_c}\frac{1}{2}(V - \hat{R})^2\|$ scales by about s; the advantages also scale by s, but are normalized per minibatch, so $\|g_a\|$ does not. By Equation 2 the applied actor gradient then falls by about $1/s$. Clipping actor and critic *separately* replaces κ with $\kappa_a = \min(1, \tau/\|g_a\|)$, which does not involve the critic, and normalizing rewards or value targets keeps $\|g_c\|$ from growing with s. Adam (Kingma & Ba, 2015) divides each coordinate by a running root-mean-square of its own gradient, so it would undo a *constant* rescaling of $\tilde{g}_a$. It does not fully undo this one, because κ varies from minibatch to minibatch with the critic's error and because the denominator is floored at ϵ. The applied gradient is therefore not the quantity that reaches the policy; the parameter update is, and it is what E6 measures directly. We do not isolate how much of the remaining attenuation is due to the varying κ and how much to ϵ.

The argument makes four testable predictions, which Section 6 checks in turn: (a) under the shared clip the critic's pre-clip norm grows with the reward multiplier while the actor's does not; (b) the actor's applied gradient, and to a lesser degree its parameter update, shrink under symmetric scaling; (c) per-group clipping restores the actor's update and removes the part of the crossing shift that runs through κ; (d) what remains after (c) runs through the critic's targets, so normalizing rewards or targets should remove it. As a quantitative check of (b), Equation 2 evaluated at the default cell's mean per-minibatch norms predicts an S/B ratio of applied actor gradients of 0.078, against a measured 0.121; because it is evaluated at averages of heavy-tailed per-minibatch norms, we read this as agreement in order of magnitude only. Figure 1 summarizes the pathway and where each intervention acts.

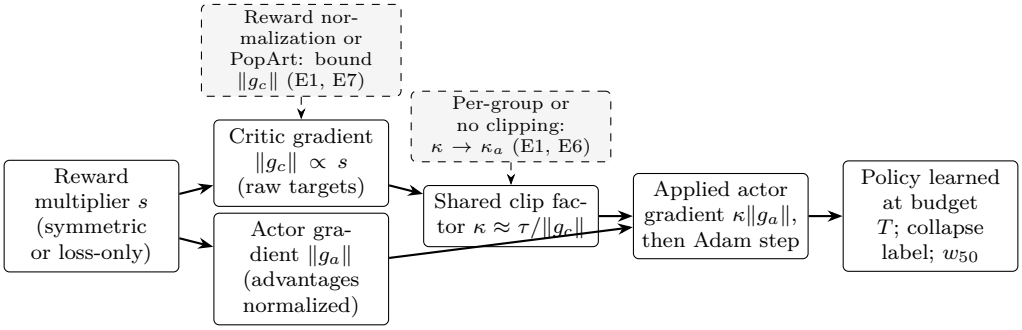


Figure 1: The shared-clip pathway. A reward multiplier enlarges the critic's gradient when value targets are not normalized, but not the actor's, because advantages are normalized per minibatch. A global clip shared by both then shrinks the actor's applied gradient roughly in proportion to $1/\|g_c\|$; Adam partly, but not fully, compensates. Dashed boxes mark where the interventions act and which experiments test them. The measured quantities are, from left to right: pre-clip norms (E1, E6), applied actor gradient and parameter update (E6), and crossing contrasts (all experiments).

5 Experimental Design and Statistical Methodology

Experiments. Table 2 lists the seven pre-registered experiments. Together they comprise 4,060 trajectories; the 2,720 in E1, E2, E6 and E7 address the optimizer mechanism, E3 and E5 follow 540 trajectories to 3.6M transitions, and E4's 800 run on the second environment. Earlier "legacy" sweeps from another platform (20 seeds per cell) supply the curvature-versus-multiplier decomposition and the cross-sectional horizon fit; they are reported separately and never pooled with E1–E7.

Estimation. We fit the logistic curve by Firth's bias-reduced maximum likelihood (Firth, 1993), which stays finite under the complete separation that ten-seed cells often show. Intervals use 2,000 multinomial

Table 1: Experimental setup. Legacy sweeps (earlier platform, 20 seeds per cell) supply the 300k decomposition and the cross-sectional horizon fit; pre-registered experiments E1–E7 run on one platform with ten seeds per cell.

	Supply chain (legacy, E1–E3)	Level-Based Foraging (E4)
Environment	Six-node `NetworkManagement-v1`; node 1 scripted, nodes 2–6 trained; 30-period episodes.	`Foraging-5x5-2p-2f-v3`; two players, two foods; 50-step episodes.
Local reward	Node i's period profit (revenue minus procurement, operating, holding, pipeline and unfulfilled-demand costs).	Native food share minus 0.02 per LOAD attempt.
Team reward	Sum of the five trained nodes' profits.	Sum of native food rewards.
Learner	Parameter-shared PPO (SB3 2.8.0); separate actor and critic MLPs.	MAPPO: actor on own observation, critic on global state (SB3 2.8.0).
Optimizer	Default: one Adam step ($\epsilon = 10^{-5}$), global clip 0.5, no value normalization. E1 adds separate no-clip optimizers and `VecNormalize`; E2 sets $\epsilon = 10^{-8}$; E6 clips actor and critic separately; E5 uses the fully decoupled learner; E7 adds a PopArt critic to the separate no-clip learner.	Global clip 0.5 or separate no-clip optimizers.
Collapse label	$\mathbf{1}[\bar{S} < 10 \wedge \bar{\pi} \leq -127.1]$ on ten deterministic evaluation episodes.	Native team return < 0.626 over ten evaluation episodes (no effort cost).
Arms	Linear; curvature α=β=.88 with $\lambda \in \{1, \ldots, 7\}$ (legacy); B, A ($\lambda = 7$), S ($\times 7$), R ($\times 3.451$) with $z_0 = -2.628$ (E1, E2); linear, curvature, $\lambda = 7$ (E3); B, A, S (E5, E6); B, A, S and an affine map M of B (E7).	Linear; curvature; A ($\lambda = 4$); S (both branches $\times 4$).
Budget, grid	300k, eight cells (E1, E2); 3.6M with 24 checkpoints, nine cells (E3, E5).	2M timesteps; ten cells $\{0, .1, .2, .3, .5, .65, .75, .85, .9, .95\}$.
Uncertainty	Firth logistic midpoint; 2,000 multinomial draws of complete training-seed blocks, shared across conditions so that contrasts are paired.	

Table 2: Experiment matrix. All experiments use ten training seeds (42–51) per arm and mixture cell, one evaluation label per run, the Firth crossing and seed-block bootstrap intervals; E4 runs on Level-Based Foraging, all others on the supply chain. E1–E4 were pre-registered together, E5–E7 each before any of its runs. "Main finding" cites the contrast that answers the row's question; Table 3 gives intervals.

ID	Question	Design (baseline → intervention)	Arms, grid, budget	Runs	Main finding	Main limitation
E1	RQ1–3: does reward scale or shape move w_{50}, by optimizer?	{shared clip, separate no-clip optimizers} × {raw, normalized rewards}	B, A, S, R; 8 cells; 300k	1,280	Default: S−B large; decoupled: S−B not detectable, A−S positive	A, S differ in reward mean
E2	Is Adam's ϵ the route?	Default with $\epsilon = 10^{-8}$	B, A, S, R; 8 cells; 300k	320	S−B unchanged	Update norms not logged
E3	Budget dependence, default learner	Default learner to 3.6M, 24 checkpoints	linear, curvature, $\lambda = 7$; 9 cells	270	Contrast shrinks; no stable crossing	Includes scale pathway
E4	Second environment	MAPPO on LBF; shared clip vs separate optimizers	linear, curvature, A, S; 10 cells; 2M	800	A−S positive in both modes	Label fallback (D4); estimator sensitivity
E5	Does A−S persist with budget?	Decoupled learner to 3.6M, 24 checkpoints	B, A, S; 9 cells	270	A−S positive at all primary budgets	Pointwise intervals
E6	Is the shared clip the pathway?	Shared vs per-group vs no clipping; update logging	B, A, S; 8 cells; 300k	720	Per-group clip restores update, reduces S−B	Residual unexplained
E7	Critic targets? Reward moments?	Plain vs PopArt critic; scale-matched control M	B, A, S, M; 8 cells; 300k	400 new + 240 E1	PopArt: inconclusive; A−M separable only with PopArt	M's mean unmatched

draws of complete training-seed blocks; the same draws serve every condition in an experiment, so contrasts are seed-paired (Efron & Tibshirani, 1993). Out-of-grid and non-converged draws are retained and counted, never clipped. Logistic, probit and decreasing isotonic fits on the same rows serve as sensitivity estimators (convention in Appendix B); their maximum spread on the supply-chain data is 0.018.

Interval calibration. The training seed is the independent unit: every interval resamples complete seed blocks, never checkpoints or evaluation episodes. With ten seeds a percentile bootstrap is known to be narrow, so we simulated the design (ten seeds, the E1 grid, one label per seed and cell) from known logistic curves, with and without a per-seed effect shared by paired arms (Appendix B, Table 14). The declared 95%

intervals covered the true crossing and the true contrast in 0.88–0.92 of 1,000 replicates per scenario, and excluded zero for a null contrast in 0.087–0.096 of them, rather than 5%. Nominal 99% intervals covered at 0.94 or more. We keep the declared 95% intervals as the primary analysis and report the 99% intervals for every principal contrast as a sensitivity check (Table 3): none of the 14 principal verdicts changes. An independent set of 10,000 bootstrap draws moves no endpoint by more than 0.007. Intervals are pointwise; we make no simultaneous claim across contrasts or checkpoints. An interval that includes zero is reported as no detectable effect, never as evidence of no effect; we make equivalence statements only against a stated margin.

Pre-registration and platform. Experiments E1–E7 were pre-declared, with source and design hashes frozen before launch; the departures (one pre-execution source fix, a label-threshold fallback corrected from stored data, and analysis-code fixes) are logged in Appendix C. All E1–E7 trajectories run on one CPU platform. An earlier platform does not reproduce trained policies bit-for-bit, but collapse labels agree in 221/240 (92.1%) of matched direct-control runs and 3,289/3,600 (91.4%) of matched horizon checkpoints, so we make reproducibility claims at the level of labels and crossings. The legacy sweeps in Sections 7 and 9 (20 seeds per cell) come from the earlier platform and are not pooled with E1–E7.

6 Symmetric Reward Scaling and the Shared Clip

Table 3 and Figure 2 collect the principal contrasts of this and the next two sections. This section tests predictions (a)–(d) of Section 4 in six steps, summarized in Table 4: an observation, a diagnostic, an intervention that isolates one pathway, a residual effect, a test of one interpretation of that residual, and the dependence of the baseline crossing on the optimizer. The quantities involved are distinct: the *pre-clip* gradient norm, the *applied* gradient after clipping, the parameter *update* after Adam, and the *crossing* estimated from evaluation behavior.

Table 3: Principal crossing contrasts. Each contrast pairs two arms over the same ten training seeds (seeds 42–51) on the experiment's mixture grid; intervals resample complete training-seed blocks (2,000 draws). The 95% interval is the pre-declared one. The 99% interval is the coverage-calibrated sensitivity of Section 5, whose simulated coverage is at least 0.94. No principal verdict differs between the two.

Exp.	Env.	Learner / intervention	Contrast	Estimate	95% interval	99% interval	Reading
E1	Supply	Shared clip, raw rewards (default)	S−B	+0.357	[+0.249, +0.464]	[+0.220, +0.495]	symmetric shift
E6	Supply	Per-group clip, raw rewards	S−B	+0.150	[+0.095, +0.226]	[+0.077, +0.245]	symmetric shift reduced
E1	Supply	Separate optimizers, normalized rewards	S−B	−0.049	[−0.117, +0.017]	[−0.139, +0.032]	none detectable
E7	Supply	Separate optimizers, PopArt critic, raw rewards	S−B	+0.075	[−0.032, +0.164]	[−0.070, +0.180]	none detectable
E1	Supply	Separate optimizers, normalized rewards	A−S	+0.237	[+0.136, +0.346]	[+0.114, +0.378]	asymmetric shift
E1	Supply	Separate optimizers, normalized rewards	A−B	+0.188	[+0.082, +0.266]	[+0.047, +0.288]	loss-only shift
E6	Supply	Per-group clip, raw rewards	A−S	+0.168	[+0.077, +0.240]	[+0.047, +0.261]	asymmetric shift
E5	Supply	Separate optimizers, normalized rewards, 3.6M	A−S	+0.154	[+0.053, +0.256]	[+0.018, +0.277]	persists at 3.6M
E7	Supply	PopArt critic, raw rewards	A−S	+0.207	[+0.141, +0.266]	[+0.126, +0.280]	asymmetric shift
E7	Supply	Plain critic, raw rewards	A−M	+0.050	[+0.000, +0.100]	[−0.012, +0.112]	not separable from moments
E7	Supply	PopArt critic, raw rewards	A−M	+0.268	[+0.211, +0.317]	[+0.194, +0.327]	exceeds scale-matched control
E4	LBF	MAPPO, shared clip	A−S	+0.447	[+0.312, +0.559]	[+0.260, +0.600]	asymmetric shift
E4	LBF	MAPPO, separate optimizers	A−S	+0.490	[+0.315, +0.619]	[+0.235, +0.670]	asymmetric shift

Design (E1, E2). The pre-registered factorial crosses two optimizer modes with two reward modes at 300k. The optimizer modes are SB3's default global clip and separate actor and critic Adam optimizers without clipping. The reward modes are raw transformed rewards and online normalization (`VecNormalize` of rewards by the running standard deviation of discounted returns, no observation normalization). Each cell runs the four arms B, A, S and R on eight mixture cells $\{0, .2, .4, .5, .6, .7, .85, 1\}$ with seeds 42–51, for

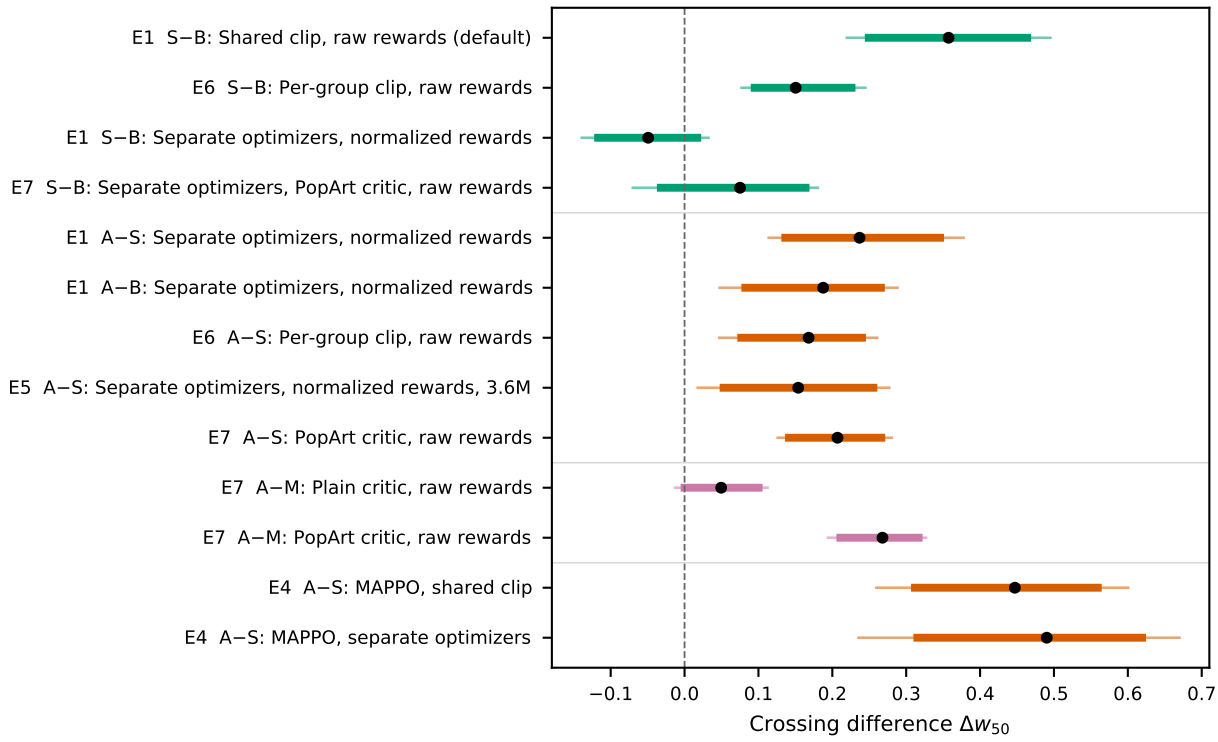


Figure 2: Principal crossing contrasts across experiments. Thick bars: pre-declared 95% seed-block intervals; thin bars: coverage-calibrated 99% intervals (Section 5); dots: point estimates. Top block: symmetric scaling S−B under the default shared clip, per-group clipping, decoupled learner with normalized rewards, and PopArt critic. Middle: asymmetric contrasts on the supply chain. Lower: asymmetric versus the scale-matched symmetric control M, and the LBF replication.

Table 4: Evidence hierarchy for the optimizer account. Each level names the experiment that supports it and how far the evidence goes. "Isolated" means an intervention changes only the named quantity; "consistent" means the data agree with the explanation but an alternative is not excluded.

Level	Finding	Evidence	Status
1. Observation	Under the default configuration, symmetric ×7 scaling moves the crossing (S−B +0.357 [+0.249, +0.464]).	E1, Table 6	Direct
2. Diagnostic	The critic dominates the shared clip: pre-clip critic norm 1,327 (B) vs. 11,965 (S); the actor's applied gradient for S is 0.12 of B's and its parameter update 0.39.	E1, E6, Tables 21 and 7	Direct
3. Intervention	Clipping actor and critic separately restores the actor update (0.95) and changes S−B by −0.207 [−0.336, −0.085].	E6	Isolated
4. Residual	Without clipping S−B is +0.160 [+0.078, +0.254]; adding reward normalization gives −0.049 [−0.117, +0.017] (+0.050 [−0.035, +0.144] under the shared clip).	E1, E6	Direct
5. Interpretation	Candidate: the residual reflects the critic's target scale. Normalizing only the critic's targets (PopArt) raises explained variance for S from 0.30 to 0.69 (B: 0.68) and changes S−B by −0.085 [−0.183, +0.006].	E7, Table 8	Tested, inconclusive
6. Moment control	A symmetric affine map of B matched to A's reward scale: A−M is +0.050 [+0.000, +0.100] with a raw-target critic and +0.268 [+0.211, +0.317] with PopArt. Its SD matches A's; its mean is more negative.	E7, Table 8	Raw targets: not separable; PopArt: direct

1,280 trajectories. An addendum (E2, 320 trajectories) keeps the default clip and raw rewards but sets Adam's ϵ to 10^{-8}. The pre-declared readings were: an A−S interval excluding zero after both interventions indicates a sign-asymmetric effect; collapse of the symmetric contrasts after an intervention indicates a scale or optimizer mechanism.

Prediction (a)–(b): under the shared clip, symmetric scaling mimics loss asymmetry. In the default cell every minibatch exceeds the clip norm, and the critic's pre-clip gradient norm grows from 1,327 (B) to 11,965 (S) while the actor's stays near 2. The applied actor update after clipping is 0.0015 for B and 0.0002 for S (Table 21). Symmetric ×7 scaling therefore moves the crossing by +0.357 [+0.249, +0.464] relative to B, at least as far as the loss multiplier, and A−S is −0.074 [−0.150, +0.000] (Table 6).

Removing the coupling removes the detectable symmetric shift. With normalized rewards the critic's gradient falls below the clip norm, the actor carries almost all of the gradient energy, and no symmetric shift is detectable (S−B +0.050 [−0.035, +0.144]). With separate optimizers alone the symmetric ×7 effect shrinks to +0.160 [+0.078, +0.254] and the ×3.451 effect is no longer detectable. With both interventions, S−B is −0.049 [−0.117, +0.017]. These intervals include zero, but they are not equivalence results: under both interventions the interval extends beyond a ±.05 margin, so a symmetric shift of that size is not excluded. The changes in each contrast relative to the default cell are in Table 20; Figure 4 summarizes them.

The artifact is not Adam's ϵ, and points to critic target magnitude. Lowering ϵ to 10^{-8} leaves the symmetric effect in place (S−B +0.258 [+0.187, +0.330]; change relative to default −0.099 [−0.243, +0.033]), contrary to our pre-declared alternative that the clipped actor gradients were being suppressed by Adam's ϵ floor. Normalizing rewards, which bounds the critic's targets, removes the detectable effect, whereas separating the optimizers only reduces it. The critic's explained variance agrees: without clipping and with raw rewards it falls from 0.68 (B) to 0.30 (S), whereas under reward normalization it is 0.67 and 0.66. Two qualifications limit this account. First, `VecNormalize` normalizes rewards, not value targets as PopArt does (van Hasselt et al., 2016); E7 below normalizes the targets alone. Second, the separate-optimizer arm

removes clipping as well; because Adam is per-parameter, splitting one Adam into two changes nothing by itself, so this arm tests the removal of clipping, not optimizer grouping.

Prediction (c): clipping isolation (E6). A second pre-registered cohort (720 runs) clips actor and critic gradients separately under one Adam step, and logs, at every optimizer step, both the applied gradient and the actual parameter change (Table 7). With logging switched on, the default and no-clip modes reproduce the corresponding E1 runs exactly (480/480 labels and profits), so the logging does not alter training. Under the shared clip the actor's applied gradient for S is 0.12 of B's, but Adam partly compensates, and the actor's actual parameter update is 0.39 of B's. With per-group clipping or no clipping the ratio is 0.95 and 0.96, close to but detectably below one, so our pre-declared prediction of no difference is not met as stated. Per-group clipping reduces the symmetric effect by −0.207 [−0.336, −0.085] to +0.150 [+0.095, +0.226], and its crossings are almost identical to those without any clipping. The shared clip therefore accounts for part of the artifact, roughly 0.2 of the 0.36 shift. The remainder is unaffected by clipping, accompanies a poorly fitted critic, and disappears under reward normalization (Section 6, E1). The asymmetric contrast A−S is +0.168 [+0.077, +0.240] under per-group clipping. Figure 3 puts the three steps side by side.

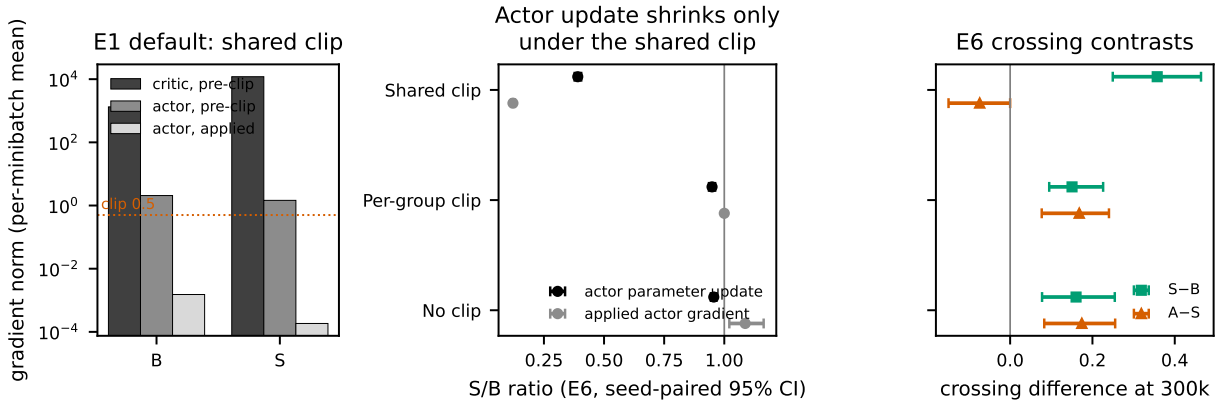


Figure 3: The shared-clip pathway. Left: in the default E1 cell (log scale) the critic's pre-clip gradient norm rises from 1,327 (B) to 11,965 (S) while the actor's stays near 2, so one global clip to 0.5 shrinks the actor's applied gradient more for S. Middle: E6 seed-paired S/B ratios of the actor's applied gradient and of its actual parameter update, with 95% intervals; a parameter-update ratio is not a gradient ratio, because Adam rescales. Right: E6 crossing contrasts. Per-group clipping restores the actor update and removes part of the symmetric shift; A−S becomes positive.

Prediction (d): critic-only target normalization (E7). Reward normalization rescales both the critic's targets and the rewards from which the actor's advantages are built. E7 keeps separate optimizers without clipping and raw rewards, and crosses two critics with four arms on the E1 grid (400 new runs; the plain-critic B, A and S runs are E1's, licensed by a gate that reproduced an E1 run bit-for-bit). The *plain* critic regresses raw returns. The *PopArt* critic reparametrizes its output as $V(x) = \sigma\, h(x) + \mu$, with μ, σ running moments of the rollout returns and an output-preserving rescale of the head whenever they change (van Hasselt et al., 2016); the actor's rewards, advantages and loss are untouched. PopArt repairs the critic's fit: explained variance is 0.68 (B) and 0.69 (S), against 0.68 and 0.30 with the plain critic. The symmetric shift falls from +0.160 [+0.078, +0.254] to +0.075 [−0.032, +0.164], but the change, −0.085 [−0.183, +0.006], does not exclude zero. Our pre-declared criterion for attributing the residual to critic target scale required both, so this reading is inconclusive: the point estimate is a partial reduction, and ten seeds cannot separate it from none (Table 8). The residual of prediction (d) is therefore removed by reward normalization but not attributed.

The baseline crossing itself depends on the optimizer. The curvature-only crossing B is 0.538 in the default cell and 0.31–0.45 in the other cells (Table 5). An absolute w_{50} is therefore a property of the

learner's optimizer configuration as well as of the reward design; comparisons are valid only within a fixed configuration.

Table 5: Optimizer × reward-normalization factorial (E1) and Adam-ϵ addendum (E2): crossings w_{50} at 300k with 95% seed-block intervals. B: curvature only; A: losses ×7; S: both branches ×7; R: both branches ×3.451 (A's pooled RMS ratio). Ten seeds per cell, eight mixture cells, one platform.

Optimizer / reward	B	A	S	R
Global clip, raw (default)	0.538 [0.461, 0.624]	0.821 [0.775, 0.864]	0.895 [0.835, 0.957]	0.794 [0.775, 0.821]
Global clip, normalized reward	0.306 [0.201, 0.413]	0.593 [0.527, 0.655]	0.356 [0.272, 0.436]	0.367 [0.324, 0.414]
Separate optimizers, raw	0.326 [0.256, 0.392]	0.661 [0.628, 0.696]	0.487 [0.412, 0.569]	0.353 [0.270, 0.442]
Separate optimizers, normalized	0.414 [0.351, 0.506]	0.602 [0.560, 0.641]	0.365 [0.282, 0.439]	0.350 [0.252, 0.438]
Global clip, raw, Adam ϵ=10^{-8}	0.450 [0.387, 0.534]	0.757 [0.729, 0.775]	0.708 [0.649, 0.770]	0.615 [0.559, 0.683]

Table 6: Paired seed-block contrasts within each optimizer cell. *: 95% interval excludes zero. A−S is the pre-declared asymmetry contrast; S−B and R−B are symmetric-scaling effects.

Optimizer / reward	A−S	A−B	S−B	R−B
Global clip, raw (default)	−0.074 [−0.150, +0.000]	+0.283* [+0.181, +0.386]	+0.357* [+0.249, +0.464]	+0.256* [+0.160, +0.344]
Global clip, normalized reward	+0.237* [+0.169, +0.305]	+0.287* [+0.199, +0.386]	+0.050 [−0.035, +0.144]	+0.060 [−0.068, +0.179]
Separate optimizers, raw	+0.174* [+0.083, +0.255]	+0.334* [+0.274, +0.405]	+0.160* [+0.078, +0.254]	+0.027 [−0.024, +0.069]
Separate optimizers, normalized	+0.237* [+0.136, +0.346]	+0.188* [+0.082, +0.266]	−0.049 [−0.117, +0.017]	−0.064 [−0.176, +0.024]
Global clip, raw, Adam ϵ=10^{-8}	+0.048 [−0.013, +0.111]	+0.307* [+0.224, +0.363]	+0.258* [+0.187, +0.330]	+0.165* [+0.109, +0.219]

Table 7: Clipping isolation (E6): one Adam step with a global clip, actor and critic clipped separately, or no clipping; raw rewards; arms B, A, S on the E1 grid; ten seeds; 720 runs at 300k. Crossing contrasts, and seed-paired S/B ratios of the actor's applied (post-clip) gradient norm and of the norm of its actual parameter update. *: interval excludes zero.

Mode	S−B	A−S	Applied actor gradient S/B	Actor update S/B
Global clip (default)	+0.357* [+0.249, +0.464]	−0.074 [−0.150, +0.000]	0.121 [0.114, 0.129]	0.391 [0.378, 0.405]
Per-group clip	+0.150* [+0.095, +0.226]	+0.168* [+0.077, +0.240]	1.000 [1.000, 1.000]	0.949 [0.936, 0.962]
No clip	+0.160* [+0.078, +0.254]	+0.174* [+0.083, +0.255]	1.087 [1.021, 1.164]	0.956 [0.944, 0.967]

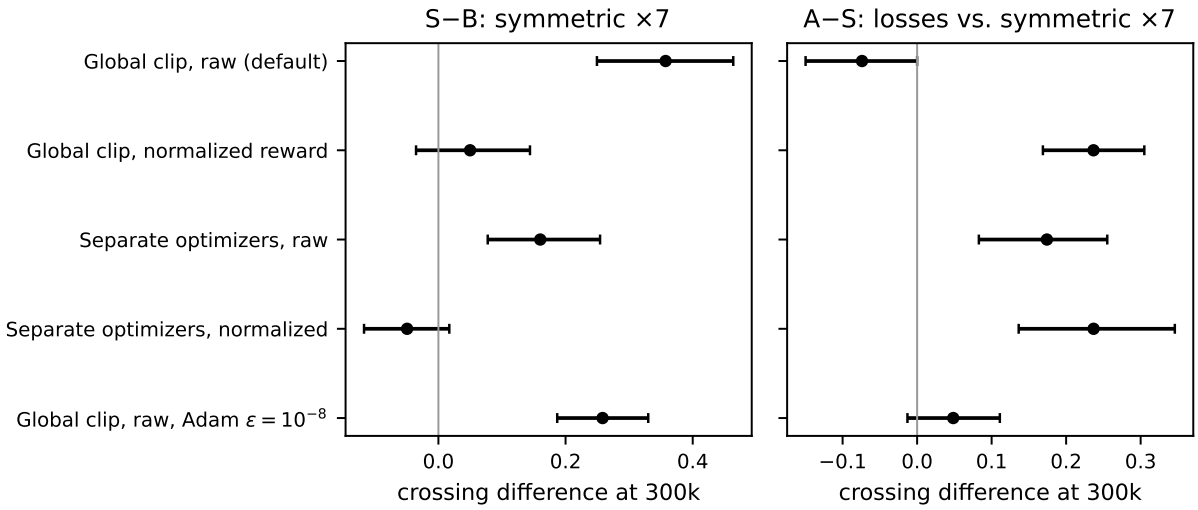


Figure 4: Symmetric-scaling (left) and asymmetric-minus-symmetric (right) crossing differences in each optimizer cell, with paired 95% seed-block intervals. Under the default configuration (top row) symmetric scaling moves the crossing as far as the loss multiplier; with normalized rewards and/or separate optimizers it does not, while A−S becomes positive.

Table 8: Critic-target normalization and a moment-matched control (E7): separate actor and critic optimizers without clipping, raw rewards, the E1 grid, ten seeds, 300k. M is the affine map $a_w r_{\mathrm{B}} + b_w$ of B's reward with constants frozen to give A's E1 training-reward mean and SD. The plain-critic B, A and S runs are E1's. Crossing contrasts with paired 95% seed-block intervals; the bottom row is the change in each contrast from the plain to the PopArt critic. *: interval excludes zero.

Critic	S−B	A−S	A−M	M−B	A−B
Plain	+0.160*	+0.174*	+0.050	+0.285*	+0.334*
	[+0.078, +0.254]	[+0.083, +0.255]	[+0.000, +0.100]	[+0.208, +0.363]	[+0.274, +0.405]
PopArt	+0.075	+0.207*	+0.268*	+0.014	+0.282*
	[−0.032, +0.164]	[+0.141, +0.266]	[+0.211, +0.317]	[−0.082, +0.113]	[+0.181, +0.380]
PopArt−plain	−0.085	+0.033	+0.218*		
	[−0.183, +0.006]	[−0.080, +0.135]	[+0.143, +0.298]		

7 Asymmetric Reward Transforms

The loss multiplier opposes curvature. In the legacy decomposition at 300k (20 seeds per cell), the linear null gives 0.685 [0.650, 0.715]; this is a separate run from the structural sweep of Section 9 (0.666 [0.631, 0.696]), with a different w-grid, and the decomposition uses its own null. Curvature alone (α=β=0.88, the conventional prospect-theory estimate (Tversky & Kahneman, 1992), not tuned) lowers the crossing to 0.585 (-0.100 [-0.145, -0.057] vs. linear), and raising λ from 1 to 7 raises it to 0.852 [0.840, 0.865] (Figure 5; per-λ values in Appendix B). Reported as a single "prospect-theoretic" treatment the two effects partially cancel. Against the curvature baseline the loss-multiplier shift is +0.268 [+0.230, +0.306]. On its own, this shift cannot distinguish loss asymmetry from the larger reward magnitude the multiplier creates; Section 6 showed how; this section separates the two.

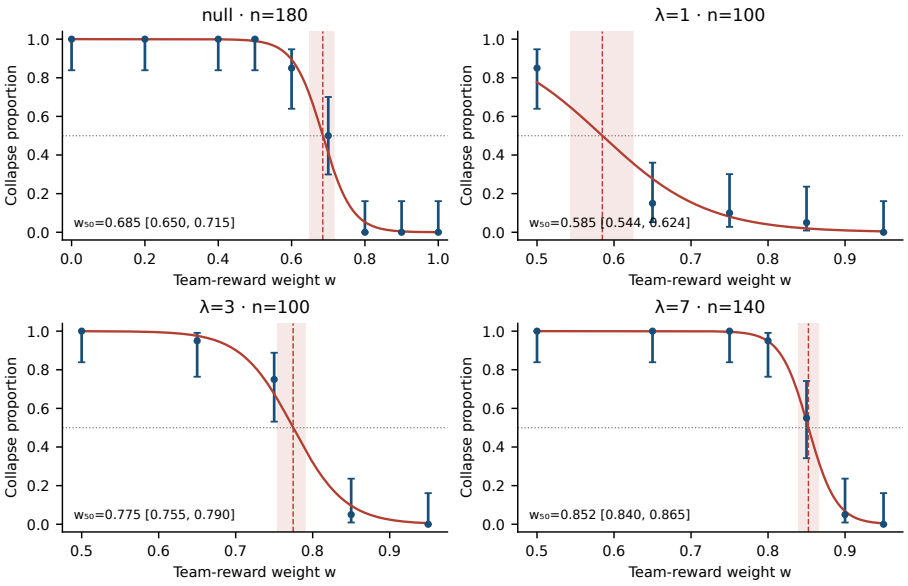


Figure 5: Legacy supply PPO null and selected transform conditions at 300k. Points show collapse counts divided by cell totals with 95% Wilson intervals; curves are Firth fits over observed grids. Vertical bands are midpoint seed-block intervals.

Equivalent linear-training budget. As a descriptive translation, inverting the legacy cross-sectional fit $w_{50} = 1.4824 - 0.13748 \ln(T/1000)$ maps curvature-only at 300k to a linear budget of 685k [535, 942] and $\lambda = 7$ to 98k [67, 128] (an extrapolation below the observed 150k–900k range); the repeated linear null maps to 330k [275, 406] (Table 9). This mapping is not causal, and it is consistent with Section 6: under the default optimizer the loss multiplier behaves like a smaller training budget.

With the coupling removed, the loss multiplier still moves the crossing (E1, E6). Because A and S apply the same multiplier to losses and differ only in whether gains are also scaled, A−S compares two gain-to-loss ratios (1:7 and 1:1) at matched loss scale, and A−B isolates the loss-branch multiplier at fixed gains. A−S is positive and excludes zero under normalization (+0.237 [+0.169, +0.305]), separate optimizers (+0.174 [+0.083, +0.255]), both together (+0.237 [+0.136, +0.346]) and per-group clipping (+0.168 [+0.077, +0.240]), and A−B is +0.188 [+0.082, +0.266] with both interventions, so the pre-declared criterion for a sign-asymmetric effect is met. How much of each reward falls on the loss branch depends on w and on the learned policy, because z_0 is fixed; near the crossings both branches are active (Table 23). The arms also differ in reward mean, not only in scale: A's transformed rewards have a negative mean at most w, whereas S and R have positive means that grow with w (Figure 12, Appendix E). A−S therefore compares gain-to-loss ratios but does not hold the first two moments fixed.

Table 9: Supply-PPO reward-transform decomposition at 300k requested agent transitions and descriptive equivalent linear-training budgets. Differences use paired seed-block draws. Equivalent budgets invert the four-point cross-sectional fit $w_{50} = 1.4824 - 0.13748 \ln(T/1000)$ and propagate the same 2,000 bootstrap replicates; they are descriptive, not causal. † denotes extrapolation below the observed 150k–900k horizon range.

Condition	Crossing and 95% CI	Versus null	Versus curvature	Equivalent budget, k
Linear null	0.685 [0.650, 0.715]	–	–	330 [275, 406]
Curvature ($\lambda = 1$)	0.585 [0.544, 0.624]	−0.100 [−0.145, −0.057]	–	685 [535, 942]
$\lambda = 3$	0.775 [0.755, 0.790]	+0.090 [+0.065, +0.115]	+0.190 [+0.150, +0.229]	172 [140, 211]
$\lambda = 7$	0.852 [0.840, 0.865]	+0.167 [+0.134, +0.203]	+0.268 [+0.230, +0.306]	98 [67, 128]†

The asymmetric contrast persists with budget (E5). To test the asymmetric effect over budget without the scale pathway, we repeated the horizon design with the fully decoupled learner of Section 6 (separate optimizers without clipping, normalized rewards) and arms B, A and S: 270 trajectories on the same nine cells to 3.6M. All three pre-declared predictions are met (Table 10). A−S is positive at every primary budget, from +0.234 [+0.165, +0.322] at 300k to +0.154 [+0.053, +0.256] at 3.6M, and no symmetric shift is detectable at any budget (S−B −0.056 [−0.147, +0.043] at 300k, −0.049 [−0.240, +0.099] at 3.6M). The loss-only contrast A−B is positive to 900k and not detectable afterwards. Under this learner the crossings fall to 900k and then rise as many initially coordinated trajectories relapse into collapse, and no arm stabilizes. The asymmetric effect therefore persists over the tested budgets while the absolute crossings keep moving. Figure 6 shows all 24 checkpoints. This reading of the non-primary checkpoints is exploratory and its intervals are pointwise. The A−S interval excludes zero at 23/24 checkpoints, with point estimates between +0.099 and +0.236. The S−B interval excludes zero at 3 intermediate checkpoints (1.05–1.5M), all with S *below* B, and at no checkpoint with S above B. The default-optimizer E3 contrast is drawn for comparison; its own crossings are in Figure 11 (Appendix D).

Table 10: Decoupled horizon cohort (E5): separate actor and critic optimizers without clipping and normalized rewards; arms B, A, S; the nine E3 cells; ten seeds; 270 trajectories to 3.6M. Crossings, then paired contrasts. *: interval excludes zero.

Arm or contrast	300k	900k	1.8M	3.6M
B (curvature)	0.425 [0.344, 0.507]	0.385 [0.332, 0.439]	0.639 [0.541, 0.715]	0.732 [0.638, 0.866]
A (losses ×7)	0.603 [0.564, 0.640]	0.526 [0.452, 0.597]	0.667 [0.612, 0.731]	0.837 [0.766, 0.916]
S (both ×7)	0.369 [0.267, 0.450]	0.329 [0.237, 0.391]	0.548 [0.427, 0.646]	0.683 [0.592, 0.759]
A−S	+0.234* [+0.165, +0.322]	+0.197* [+0.086, +0.328]	+0.119* [+0.034, +0.244]	+0.154* [+0.053, +0.256]
A−B	+0.178* [+0.081, +0.271]	+0.141* [+0.064, +0.224]	+0.028 [−0.044, +0.113]	+0.105 [−0.045, +0.242]
S−B	−0.056 [−0.147, +0.043]	−0.056 [−0.146, +0.016]	−0.091 [−0.262, +0.054]	−0.049 [−0.240, +0.099]

Against a scale-matched symmetric control, the effect depends on the critic (E7). E7's arm M is an increasing affine map $a_w r_B + b_w$ of B's reward, symmetric in gains and losses, with constants frozen so that M has A's E1 training-reward mean and SD. With the plain critic, M moves the crossing by +0.285 [+0.208, +0.363] relative to B, most of A's +0.334 [+0.274, +0.405], and A−M is +0.050 [+0.000, +0.100]; its lower bound touches zero ([−0.012, +0.112] at the calibrated level), so the pre-declared prediction that A exceeds M under both critics is not met. Under a critic trained on raw targets, therefore, the A−S contrast may be an effect of the reward's moments rather than of its sign asymmetry. With the PopArt critic M is indistinguishable from B (+0.014 [−0.082, +0.113]), while A exceeds both S (+0.207 [+0.141, +0.266]) and M (+0.268 [+0.211, +0.317]); the change in A−M between critics is +0.218 [+0.143, +0.298].

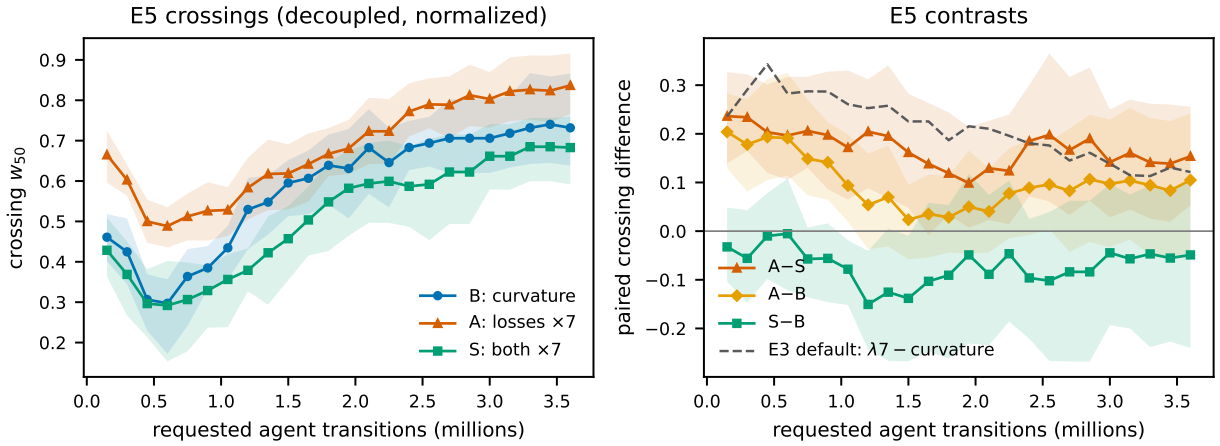


Figure 6: Decoupled horizon cohort (E5) at all 24 checkpoints. Left: crossings of B, A and S with pointwise 95% seed-block bands. Right: paired contrasts with bands; the dashed line is the default-optimizer E3 contrast $\lambda 7-$curvature (the E3 analogue of A−B, which includes the scale pathway). Nine mixture cells, ten seeds per cell and arm.

One qualification applies to both readings (Figure 7). The affine constants match A's moments under E1's policies, but M's achieved moments depend on its own: M matches A's SD (32.2 vs. 30.1 with the plain critic, 30.4 vs. 29.5 with PopArt) but not its mean (-13.5 vs. -6.6; -4.8 vs. -3.1), failing the pre-declared 25% matching check. The mismatch is concentrated at $w \in [0.4, 0.7]$. There M's policies, like A's, mostly collapse, whereas the B policies whose moments set M's constants mostly do not, and B's reward is lower under a collapsed policy. M is thus a scale-matched control whose mean is more negative than A's, and the plain-critic reading cannot separate a moment effect from that mean difference. An affine control can match at most the first two moments; what it leaves different, by construction, is the shape of the reward distribution, which is the feature an asymmetric transform is meant to change. The training-reward logs confirm this and show one further difference (Table 24, Appendix E). A's per-rollout reward is left-skewed (quantile skewness -0.12 (plain) and -0.27 (PopArt)), M's is not ($+0.18$ and -0.02). With the plain critic, M's critic targets are also larger than A's (mean return -185 against -104, SD 148 against 113). Larger target scale is the pathway with which the residual symmetric shift is consistent (Section 6), so with a raw-target critic M is exposed to it at least as much as A. This is consistent with, but does not establish, a target-scale explanation of the near-null plain-critic A−M. A control whose moments track A's under its own policy is pre-registered as an outstanding experiment (Section 10).

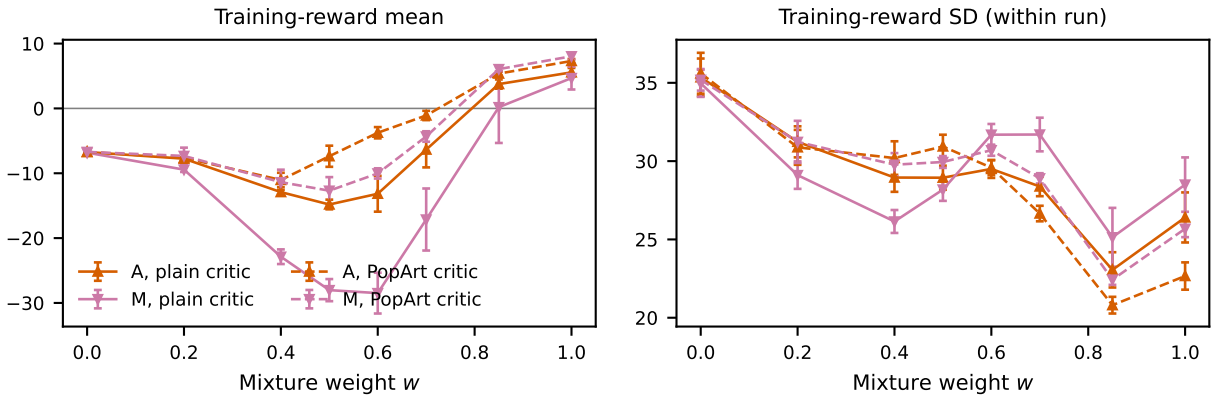


Figure 7: Achieved training-reward moments of the loss-multiplier arm A and the scale-matched symmetric control M in E7, by mixture weight (mean over ten seeds; bars are ±1.96 standard errors across seeds). M matches A's within-run SD closely but its mean falls below A's at middle weights, most with the plain critic: there M's policies mostly collapse, unlike the B policies whose moments set its constants. The plain-critic A runs are E1's.

8 A Second Environment: Level-Based Foraging

Design (E4). We repeated the arm comparison with centralized-critic MAPPO (Yu et al., 2022) on Level-Based Foraging (Christianos et al., 2020) (5×5 grid, two players, two foods), using the same SB3 update with a global clip or with separate optimizers. The local reward is the native food share minus an effort cost of 0.02 per LOAD attempt; the team reward sums native food rewards; collapse is a native team return below $\theta = 0.626$ over ten evaluation episodes, the midpoint between collapsed and coordinated team returns in an earlier calibration; the evaluation never includes the effort cost. The arms are linear, curvature only, A ($\lambda = 4$) and S (both branches ×4), on ten mixture cells with ten seeds and 2M timesteps (800 trajectories).

The asymmetric contrast replicates under both optimizers. A raises the crossing relative to S by +0.447 [+0.312, +0.559] with the global clip and +0.490 [+0.315, +0.619] with separate optimizers, and relative to curvature only by +0.543 [+0.404, +0.638] and +0.617 [+0.394, +0.767] (Table 11); the two optimizer modes do not differ detectably. The asymmetric contrast holds for collapse thresholds from 0.5 to 0.75 and under the isotonic estimator (Table 26). Here, unlike the supply chain, the critic does not dominate the clip: the actor carries 0.83–0.98 of the squared gradient norm in every arm. The loss branch is reached almost only through the LOAD cost, so the loss multiplier changes the cost of the cooperative action relative to food rewards, while S scales both and leaves their ratio unchanged; under A the share of training rewards below zero falls to a few percent at low and middle w (Table 27), because agents stop attempting loads. We therefore read the LBF result as a second environment in which an asymmetric transform moves the crossing through an incentive change, not as an independent confirmation of the supply-chain mechanism. Figure 8 shows the observed proportions behind Table 11. Several arms are far from the logistic shape, which is why Firth and isotonic crossings differ more here than on the supply chain (by up to 0.161). Figure 9 shows every arm's crossing under both estimators and three label thresholds: A's crossing lies above S's in all twelve combinations, whereas the linear arm's crossing under the global clip leaves the grid and the curvature arm's moves with the threshold, so only the A−S comparison is stable to these choices. The LBF label is a different quantity from the supply-chain label, so LBF and supply-chain crossings are compared only in sign and ordering, never in magnitude.

Two pre-declared predictions are not confirmed. We had predicted that curvature would raise the LBF crossing under the global clip (P1) and that separate optimizers would shrink the curvature and symmetric shifts (P3). Under the global clip the linear arm's crossing lies below the sampled grid (Table 11), so P1 and the curvature half of P3 have positive point estimates but are not identified; the symmetric half of

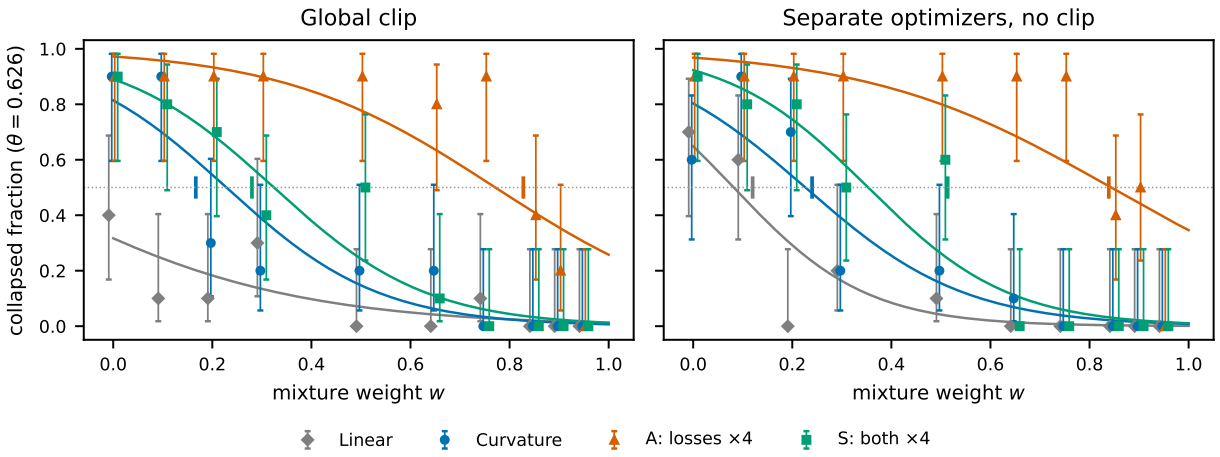


Figure 8: E4 (MAPPO on LBF): observed collapse proportions at $\theta = 0.626$ with 95% Wilson intervals (ten seeds per point; arms offset slightly in w), Firth fits (lines) and decreasing-isotonic crossings (vertical ticks at 0.5). The global-clip linear arm has no isotonic crossing in the grid and its Firth crossing lies below it (Table 11).

P3 is not met. Both predictions assumed the supply-chain clipping pathway, which LBF does not exhibit. The stored run labels used a placeholder threshold of 0.5 instead of the declared 0.626; all E4 results here are recomputed at 0.626 from the stored team returns (deviation D4, Appendix C).

Table 11: MAPPO on Level-Based Foraging (E4) at the pre-declared label $\theta = 0.626$: crossings and paired contrasts. Ten seeds, ten mixture cells, 2M timesteps, 800 trajectories. A−S compares the gain-to-loss ratio at matched loss scale; A−curvature isolates the loss-branch multiplier. †: point estimate outside the sampled grid (the linear crossing under the global clip), so contrasts that use it are not identified. *: identified contrast whose interval excludes zero. Threshold sensitivity is in Table 26.

Optimizer	Linear	Curvature	A	S	A−S	A−curvature
Global clip	$-0.210^{\dagger}$ [−0.809, 0.031]	0.228 [0.133, 0.334]	0.771 [0.584, 0.876]	0.324 [0.179, 0.451]	$+0.447^{*}$ [+0.312, +0.559]	$+0.543^{*}$ [+0.404, +0.638]
Separate, no clip	0.082 [−0.045, 0.169]	0.226 [0.117, 0.326]	0.843 [0.623, 0.918]	0.353 [0.224, 0.468]	$+0.490^{*}$ [+0.315, +0.619]	$+0.617^{*}$ [+0.394, +0.767]

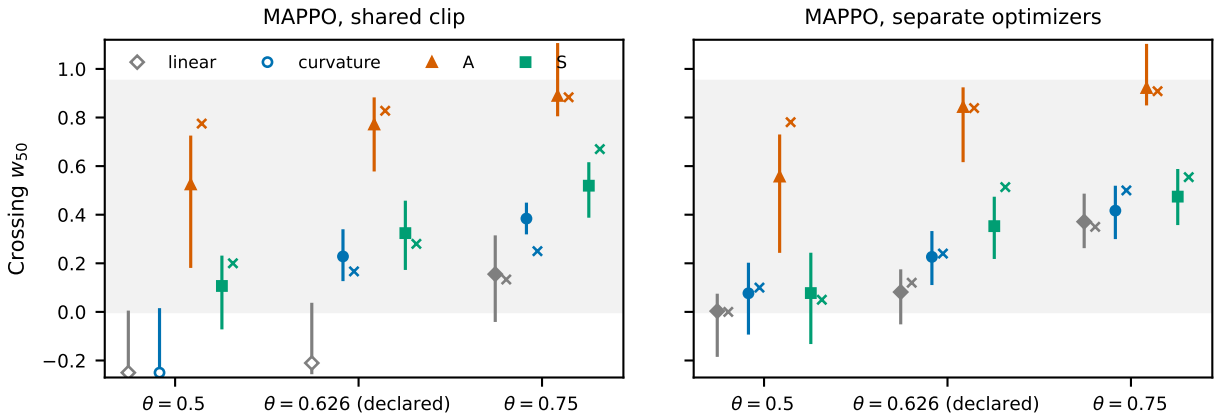


Figure 9: E4 estimator and label sensitivity. Firth crossings (markers, with 95% seed-block intervals; open markers lie outside the sampled grid, which is shaded) and decreasing-isotonic crossings (crosses) for each arm, at the declared label threshold and two alternatives. Values outside $[-0.25, 1.1]$ are drawn at the edge. The declared analysis is the middle group of each panel.

9 Robustness and Sensitivity

A well-defined boundary exists at fixed training budget. At 300k steps on the supply-chain PPO substrate, collapse has a clear, tightly-estimated boundary in reward-weighting space (Figure 10): the midpoint is 0.666 $[0.631, 0.696]$. Observed proportions and Wilson intervals show the sampled support of the crossing.

"Collapse" is an operational threshold crossing, not a confirmed attractor. We tested the continuous base-stock statistic $\bar{S}$ for a distinct dynamical regime with bootstrap 95% CIs on Sarle's bimodality coefficient (Pfister et al., 2013) and Hartigan's dip test (Hartigan & Hartigan, 1985) in each cell of the legacy curvature-only sweep ($\lambda = 1$, α=β=0.88; $w \in \{.5, .65, .75, .85, .95\}$; $n = 20$ seeds per cell). Sarle's coefficient rises near the crossing (0.59 at w=0.65 and 0.57 at w=0.75, both above the 0.555 rule of thumb) but its CI straddles the threshold everywhere, and the dip test never rejects unimodality ($p \geq 0.30$). At $n = 20$ the dip test, calibrated against the uniform least-favorable null, has little power, so non-rejection is not evidence of unimodality; it only means the data do not establish a second regime. We therefore use *threshold-defined behavioral crossing* throughout.

The boundary is finite-horizon. In the legacy cross-sectional sweep the linear-reward boundary slides left with budget: 0.821, 0.666, 0.622 and 0.572 at 150k, 300k, 450k and 900k requested agent transitions (Figure 10). These are separate finite-budget estimates and say nothing about within-policy recovery.

Refined single-platform cohort (E3). To follow the same policies, we trained 270 trajectories to 3.6M transitions, evaluated every 150k: linear, curvature-only and $\lambda = 7$ rewards, ten seeds, and nine mixture cells $\{.2, .4, .6, .65, .7, .75, .8, .85, 1\}$. The four cells between .6 and .85 were added because a coarser five-cell grid gave zero-width bootstrap intervals there. E3 uses the *default* optimizer configuration (single Adam step, global clip, raw rewards). Section 6 shows that in this configuration reward scale alone moves the crossing, so the E3 contrasts below include that scale pathway: they measure how transform effects evolve with budget under the default learner, not loss asymmetry. The multiplier-minus-curvature difference is positive at every primary budget but shrinks, from +0.291 [+0.233, +0.341] at 300k to +0.121 [+0.064, +0.177] at 3.6M (Table 12, Figure 11). The $\lambda = 7$ crossing falls monotonically, whereas the linear and curvature-only crossings fall until 900k and then rise again. No condition meets the pre-declared stability criterion, a 3.6M−1.8M interval within ±.05 (Table 13), so all readings are budget-conditional.

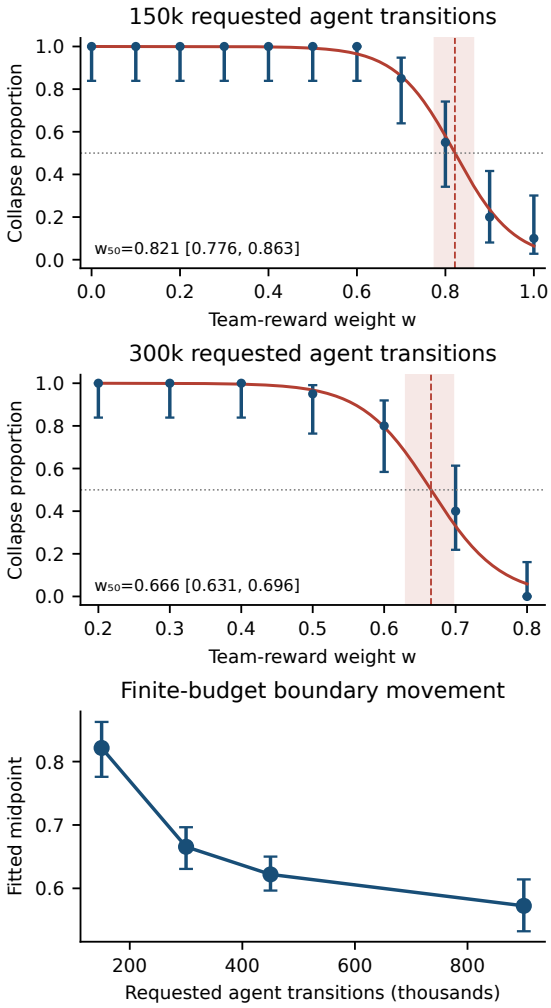


Figure 10: Finite-budget supply PPO crossings in the legacy sweep. Top and middle: observed collapse proportions with 95% Wilson intervals, Firth curves drawn only over sampled grids, midpoint lines and seed-block 95% CI bands. Bottom: independently fitted horizons; budgets count requested agent transitions.

Collapse is not absorbing, and neither is non-collapse. Conditioning on the 300k label, many collapsed trajectories recover by 3.6M and some initially non-collapsed ones relapse (Table 13). The non-monotone linear and curvature crossings reflect this relapse at high w. Cell-level counts are in Appendix D.

Table 12: Refined single-platform horizon cohort (E3), default optimizer configuration (global clip, raw rewards): nine mixture cells $\{.2, .4, .6, .65, .7, .75, .8, .85, 1\}$, ten seeds, 270 trajectories trained to 3.6M and evaluated every 150k. Crossings with 95% seed-block intervals; last row is the paired contrast.

Condition	300k	900k	1.8M	3.6M
Linear	0.663	0.512	0.535	0.578
	[0.628, 0.698]	[0.449, 0.584]	[0.471, 0.599]	[0.515, 0.643]
Curvature only	0.587	0.473	0.523	0.559
	[0.540, 0.629]	[0.400, 0.558]	[0.453, 0.597]	[0.513, 0.612]
$\lambda = 7$	0.877	0.760	0.710	0.681
	[0.843, 0.909]	[0.745, 0.775]	[0.690, 0.730]	[0.634, 0.713]
$\lambda 7-$curvature	$+0.291^*$	$+0.287^*$	$+0.187^*$	$+0.121^*$
	[+0.233, +0.341]	[+0.193, +0.364]	[+0.123, +0.247]	[+0.064, +0.177]

Table 13: Late-budget change and within-policy outcomes (E3). Stability requires the 3.6M−1.8M interval to lie within ±.05 (pre-declared for v4). Recovery: collapsed at 300k, not collapsed at 3.6M; relapse: the reverse.

Condition	3.6M−1.8M	Stable	Recovered	Relapsed
Linear	+0.043	no	14/39	8/51
	[−0.008, +0.095]			
Curvature only	+0.036	no	4/28	4/62
	[−0.029, +0.105]			
$\lambda = 7$	−0.029	no	34/76	0/14
	[−0.071, +0.001]			

Estimators, seeds and intervals. Logistic, probit and isotonic crossings agree within 0.018 on the supply chain (Appendix B); on LBF they differ more, but not in the A−S ordering (Figure 9). Leaving out any one of the ten training seeds changes the sign of none of the principal contrasts (Table 16). An independent 10,000-draw bootstrap reproduces every principal interval to within 0.007, and a coverage-calibrated 99% interval changes none of the 14 principal verdicts (Table 3). Stored labels are recomputed from the stored evaluation statistics by one module that checks each threshold against the pre-registration text (Appendix H).

10 Limitations

Mechanism. The shared-clip part of the symmetric shift is isolated by an intervention (E6). The remainder is removed by reward normalization, but a critic-only target normalization (PopArt) shrinks it by an amount ten seeds cannot separate from zero, so the critic-target-scale pathway is neither confirmed nor ruled out. How much of Adam's partial compensation is due to the varying clip factor and how much to ϵ is not isolated.

Reward moments. A and S differ in the training reward's mean and variance as well as in the gain-to-loss ratio. Against a symmetric control matched to A's reward scale, A's effect is not separable from a moment effect under a raw-target critic, and is separable under a PopArt critic; the control matched A's SD but not its mean. *Outstanding experiment.* An online moment-matched control, whose affine constants are re-set every rollout so that its training-reward mean and SD track A's under its own policy, together with an independent ten-seed replication of the PopArt comparison, is pre-registered (480 trajectories, `PREREGISTRATION_E8.md` in the artifact) but has not been run; no result from it is reported.

Statistics. Ten training seeds per cell give contrast intervals of about ±0.1 at 300k and up to ±0.17 at 3.6M, and the declared 95% intervals under-cover at this sample size (Section 5). Intervals that include

zero are reported as no detectable effect, and none would support equivalence at a ±0.05 margin. The 24-checkpoint E5 reading is exploratory and its intervals are pointwise.

Information structure and generality. The supply-chain policy is a fully observed shared controller without agent identity (Section 3); the results say nothing about decentralized information. Two small environments and one PPO implementation (SB3 2.8.0) are tested; other implementations, clipping conventions and value normalizations may couple actor and critic differently, and LBF does not exhibit the supply-chain clipping pathway. In LBF the linear arm's crossing under the global clip lies below the sampled grid, so contrasts with it are not identified. No condition has a stable crossing by 3.6M, so all readings are finite-budget. An earlier ablation on the supply chain found no interior crossing for PPO, QMIX or IQL on a coarse 11-level discretization of the base-stock action. Crossings are conditional on the behavioral label and the sampled grid; non-bracketed estimates are reported, not clipped.

Pre-registration record. E4's stored labels used a fallback threshold of 0.5 instead of the declared 0.626 because a calibration file was missing from the frozen source; all E4 results are recomputed from the stored team returns (deviation D4). Each stored team return equals the mean of the run's ten stored per-episode returns (largest difference 2×10^{-16}, all 800 runs), and the stored labels are exactly the 0.5-threshold labels. 1 run has a return of exactly 0.75, so its label in the $\theta = 0.75$ sensitivity analysis depends on floating-point summation order; no other threshold is affected. The provenance of 0.626 is an earlier calibration whose raw records are not archived, so the threshold itself is taken as declared, not re-derived; Figure 9 shows that the A−S ordering does not depend on it.

11 Discussion and Conclusion

What is established. Three findings are supported at the level of evidence stated in Table 4 and Sections 7 and 8. First, a finite-budget crossing along the reward-weighting dial can be estimated with seed-paired intervals of roughly ±0.03–0.1, but it moves with training budget, does not stabilize by 3.6M transitions, and depends on the optimizer configuration. Second, under SB3's default PPO a symmetric reward multiplier moves the crossing at least as far as a loss-only multiplier, and the shared gradient clip is the identified pathway for part of it: the critic takes the clip, the actor's parameter update shrinks, and clipping the two separately restores the update and reduces the shift. Third, with separate optimizers and normalized rewards, a loss-only multiplier moves the crossing relative to a symmetric control on the supply chain, through 3.6M transitions, and an asymmetric transform moves it on Level-Based Foraging by a different route. On the supply chain this is established against a scale-matched control only when the critic's targets are normalized; with raw targets, a symmetric control of the same scale moves the crossing nearly as far (E7). None of this is evidence about loss aversion as a psychological mechanism: the effect is a property of the transformed training signal, and in LBF it is plainly an incentive change on a costly cooperative action.

Implications for reward-transform experiments. The practical lesson is about design, not about loss asymmetry. A reward-transform comparison in actor–critic MARL should include a symmetric-scaling control at the same multiplier and should report whether actor and critic share a gradient clip and whether rewards or value targets are normalized; without these, a change in reward scale can pass for a change in reward shape.

Implications for actor–critic optimization. The confound needs two ingredients that are SB3 PPO's defaults: an unnormalized critic and one gradient clip over actor and critic. Advantages are normalized per minibatch, so the actor's loss is nearly scale-free, but the critic's gradient grows with the reward and takes the clip budget. Cooperative PPO implementations that normalize value targets by default (Yu et al., 2022) should be less exposed; we did not test them.

Interpreting finite-budget crossings. w_{50} is a 50% point of a fitted response curve at a stated budget, learner, reward unit and label. It is not a critical point, a safe operating point or an asymptotic property. Trajectories both recover from and relapse into collapse (Table 13), so the crossing summarizes a population of training outcomes, not an attractor. Contrasts between arms within one configuration are more stable than

absolute crossings: in E5 the absolute crossings fall and rise again, while A−S stays positive. A deployment requirement such as at most 5% collapse would need its own probability target and uncertainty assessment.

Reproducibility. All E1–E7 runs share one platform; trained policies are not bit-reproducible across hardware and software builds, but labels agree in 221/240 (92.1%) of matched runs, so we claim reproducibility of labels and crossings, not of policies. The artifact rebuilds every reported table, figure and number from per-run records with one command (Appendix H).

Conclusion. Reward centralization is usually treated as a direction: more team signal helps. It also has a measurable finite-budget location, w_{50}, that moves with reward transforms, training budget and the optimizer. Measured with default SB3 PPO, a symmetric reward multiplier and a loss-only multiplier cannot be told apart, because a shared gradient clip lets reward scale reach the actor's updates. With that coupling removed and rewards normalized, the loss-only multiplier moves the crossing and the symmetric one produces no detectable shift, in the tested configurations of two environments. Whether that shift is about sign asymmetry or about the reward's moments depends on the critic: with raw rewards and raw value targets, a symmetric control of matched scale moves the crossing nearly as far; with normalized value targets, it does not. Reward-transform comparisons in actor–critic MARL should therefore include symmetric-scaling controls and report the optimizer coupling, gradient clipping, reward or value normalization and training budget under which they were run.

Ethical Statement

This work is entirely simulated, with no human subjects or personal data. Reported thresholds are substrate-specific and must be re-measured before use.

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A Experimental Configuration

Supply-chain environment and learner. The six-node `NetworkManagement-v1` network has 30-period episodes. Node 1 is scripted to order 20 units per period, split equally across its action entries; nodes 2–6 are controlled by one parameter-shared PPO policy. Every trainable node's lane receives the same full or-gym observation of the network state, with no agent identifier; actions are padded to the largest per-node action dimension, and each node uses its leading entries, so under deterministic evaluation all five nodes receive the same action vector. PPO uses Stable-Baselines3 2.8.0 with `n_steps` = 128, minibatch size 256, ten optimization epochs, learning rate 3×10^{-4}, discount 0.99, GAE coefficient 0.95, policy clip 0.2, value-loss coefficient 0.5, zero entropy coefficient, normalized minibatch advantages, separate actor and critic MLPs of width 64, one Adam optimizer with SB3's implicit $\epsilon = 10^{-5}$, and global gradient max-norm 0.5. Training is CPU-only and single-threaded, with an explicit simulator random stream per environment.

Budget arithmetic. Budgets count requested agent transitions. Five lanes and 128-step rollouts add 640 transitions per rollout, and training stops at the first completed rollout at or beyond the request: 300k requested transitions execute 300,160 (469 rollouts), and each 150k checkpoint is evaluated at the first rollout boundary after it. One agent transition is one node's step, so 300,160 transitions are 60,032 joint environment periods (about 2,001 episodes).

Reward, reference point and controls. Each agent receives $x_i(w) = (1-w)r_i^{\text{local}} + wr^{\text{global}}$ before any transform. The linear condition uses $v(x) = x$; curvature-only uses $\phi(x) = \text{sign}(x)|x|^{.88}$; the loss-multiplier condition uses $v_\lambda(x) = \phi(x)$ for $x \geq 0$ and $\lambda\phi(x)$ for $x < 0$. E1 and E2 use the reference $z_0 = -2.6281398333$, fixed before any of their runs as the pooled mean of raw blended rewards from 40 untreated curvature-only calibration policies, with 7,500 samples at each of the eight E1 mixture cells (60,000 samples). With $u = x - z_0$ and $s_{\text{in}} = 1$, the arms are B= $\phi(u)$ ($s_{\text{out}} = 1$), A= $v_7(u)$ ($s_{\text{out}} = 1$), S= $7\phi(u)$ ($s_{\text{out}} = 1/7$) and R= $3.451\,\phi(u)$ ($s_{\text{out}} = 1/3.451$, where 3.451 is A's pooled transformed-reward RMS divided by B's in the earlier direct control). E3 uses reference zero.

Optimizer interventions. `separate_no_clip` builds two Adam optimizers with the default settings, one over the actor parameters (policy MLP, action head, log-standard-deviation) and one over the critic parameters (value MLP and head), applies the unchanged PPO loss, and skips gradient clipping; execution aborts if any trainable parameter is shared. `VecNormalize` divides rewards by the running standard deviation of discounted returns ($\gamma = 0.99$, $\epsilon = 10^{-8}$, reward clip 10, no observation normalization). E2 passes `optimizer_kwargs={eps: 1e-8}`. E6's `groupwise_clip` keeps one Adam optimizer and clips the actor and critic gradients to norm 0.5 separately; E6 also records, at every optimizer step and for each parameter group, the post-clip gradient norm and the norm of the parameter change, by copying the parameters before the step. Every run logs per-minibatch actor, critic and total gradient norms before and after the clipping step.

Labels. The supply-chain label is $\mathbf{1}[\bar{S} < 10 \wedge \bar{\pi} \leq -127.1]$, with $\bar{S}$ the mean over nodes 2–6 of the deterministic base-stock target $250(a_1 + 1)$ at the reset of evaluation seed $10s$, and $\bar{\pi}$ the ten-episode mean total profit on seeds $10s, \ldots, 10s+9$. The LBF label is native team return below 0.626 over ten deterministic episodes on seeds $10s, \ldots, 10s + 9$; the native return excludes the effort cost.

LBF learner. MAPPO uses SB3 PPO with a centralized-critic policy: the actor's features come from the agent's own observation and the critic's from the concatenated global state (feature width 128; actor heads $[64, 64]$, critic heads $[128, 128]$), eight parallel environment copies, `n_steps` = 256, minibatch 512, entropy coefficient 0.01, and `lbforaging` 2.0.0. In a check at 100k timesteps on one cell and seed, the global-clip runner reproduced the legacy LBF runner's team return exactly.

B Boundary Estimation and Diagnostics

For each fixed configuration we fit $P(Y = 1 \mid w) = \sigma(b_0 + b_1 w)$ by Firth logistic regression and report $w_{50} = -b_0/b_1$. A point estimate is reported only with an identified nonzero slope; observed proportions

are inspected separately for adjacent cells bracketing .5. Confidence intervals use 2,000 percentile-bootstrap draws of complete training-seed blocks (multinomial weights over the ten seeds; random seed 20261015 for E1–E7). The same weight vector is used for every condition in an experiment, so differences are seed-paired. Fits outside the grid and non-converged draws are retained and counted. Intervals are pointwise; no simultaneous statement across checkpoints or cells is made.

Interval coverage and bootstrap stability. These checks were added in revision; they are not pre-registered and change no reported estimate (`analysis/stat_checks.py`). First, each principal contrast is re-derived from the per-run tables and asserted equal to the stored result. Second, an independent set of 10,000 draws (seed 20261105) moves no interval endpoint by more than 0.007. Third, Table 14 simulates the design from two logistic curves, steep ($b_1 = -20$) and shallow ($b_1 = -8$), as found in the E1 and E7 fits, with a null contrast or a contrast of the size observed, with and without a per-seed logit intercept (SD 1) shared by both arms. The percentile intervals under-cover at ten seeds, as percentile bootstraps do at small sample sizes; nominal 99% intervals reach at least 0.94. Table 3 reports both levels.

Table 14: Simulated behaviour of the seed-block percentile intervals at the paper's design: ten seeds, the E1 grid, one label per seed and cell, 1,000 replicates per row, 2,000 bootstrap draws. Each scenario has two arms; "seed SD" is the standard deviation of a per-seed logit intercept shared by both arms. Coverage is of the true crossing (w_{50}) and of the true paired contrast; the last block is the fraction of replicates whose contrast interval excludes zero, which is the false-positive rate under a null contrast and power otherwise.

				Coverage of w_{50}			Coverage of contrast			Interval excludes 0		
Scenario	Seed SD	True w_{50}	True contrast	95	97.5	99	95	97.5	99	95	97.5	99
steep, null	0	0.400	+0.00	0.923	0.942	0.963	0.913	0.940	0.955	0.087	0.060	0.045
steep, null	1	0.400	+0.00	0.917	0.944	0.971	0.904	0.925	0.948	0.096	0.075	0.052
steep, +0.10	0	0.400	+0.10	0.919	0.944	0.966	0.899	0.938	0.961	0.822	0.759	0.670
steep, +0.10	1	0.400	+0.10	0.922	0.947	0.966	0.911	0.941	0.961	0.806	0.751	0.659
shallow, null	0	0.438	+0.00	0.890	0.931	0.950	0.905	0.937	0.962	0.095	0.063	0.038
shallow, null	1	0.438	+0.00	0.889	0.929	0.962	0.912	0.940	0.961	0.088	0.060	0.039
shallow, +0.20	0	0.438	+0.20	0.884	0.915	0.939	0.900	0.927	0.947	0.944	0.909	0.849
shallow, +0.20	1	0.438	+0.20	0.912	0.937	0.960	0.904	0.936	0.961	0.934	0.898	0.841

Logistic, probit and decreasing isotonic fits are sensitivity estimators on the same rows. The isotonic fit is weighted pool-adjacent-violators; its crossing is linear interpolation between the last cell with fitted probability at least .5 and the first below it, and a plateau at exactly .5 is reported as an interval with its midpoint used for comparison. The maximum spread among the three point estimators on the legacy supply-chain sweep is 0.018. Table 15 compares Firth and isotonic crossings for every in-grid E1–E4 fit.

Table 15: Agreement between Firth and decreasing-isotonic crossings for every in-grid fit. LBF collapse proportions are far from the logistic shape for some arms, which is why E4 shows the largest gaps; E4's A−S conclusion holds under both estimators (Table 26).

Experiment	Max |Firth−isotonic|	Median
E1/E2 (20 fits)	0.055	0.019
E3 (72 fits)	0.047	0.016
E4, $\theta = 0.626$ (7 in-grid fits)	0.161	0.044

Table 16: Leave-one-seed-out sensitivity of the principal contrasts (exploratory). Each contrast is refitted ten times on nine training seeds; the range of the ten point estimates and the number that keep the full-data sign are shown. E4 uses the pre-declared $\theta = 0.626$.

Experiment	Contrast	All seeds	Leave-one-out range	Same sign
E1 default	S−B	+0.357	[+0.325, +0.384]	10/10
E1 default	A−S	−0.074	[−0.098, −0.048]	10/10
E1 decoupled	S−B	−0.049	[−0.072, −0.031]	10/10
E1 decoupled	A−S	+0.237	[+0.200, +0.267]	10/10
E5, 3.6M	S−B	−0.049	[−0.102, −0.013]	10/10
E5, 3.6M	A−S	+0.154	[+0.134, +0.189]	10/10
E6 per-group clip	S−B	+0.150	[+0.135, +0.169]	10/10
E6 per-group clip	A−S	+0.168	[+0.148, +0.193]	10/10
E4 global clip	A−S	+0.447	[+0.423, +0.504]	10/10
E4 separate	A−S	+0.490	[+0.453, +0.533]	10/10

Table 17: Observed supply PPO collapse counts at 300k. Entries are $w : k/n$; Wilson intervals and all-setting cell records are exported in collapse_cells.csv.

Condition	Observed cells
null	0.00: 20/20, 0.20: 20/20, 0.40: 20/20, 0.50: 20/20, 0.60: 17/20, 0.70: 10/20, 0.80: 0/20, 0.90: 0/20, 1.00: 0/20
λ =1	0.50: 17/20, 0.65: 3/20, 0.75: 2/20, 0.85: 1/20, 0.95: 0/20
λ =2	0.50: 20/20, 0.65: 17/20, 0.75: 6/20, 0.85: 0/20, 0.95: 0/20
λ =3	0.50: 20/20, 0.65: 19/20, 0.75: 15/20, 0.85: 1/20, 0.95: 0/20
λ =4	0.50: 20/20, 0.65: 20/20, 0.75: 18/20, 0.85: 3/20, 0.95: 0/20
λ =5	0.50: 20/20, 0.65: 20/20, 0.75: 20/20, 0.80: 17/20, 0.85: 3/20, 0.90: 0/20, 0.95: 0/20
λ =6	0.50: 20/20, 0.65: 20/20, 0.75: 17/20, 0.80: 17/20, 0.85: 10/20, 0.90: 0/20, 0.95: 0/20
λ =7	0.50: 20/20, 0.65: 20/20, 0.75: 20/20, 0.80: 19/20, 0.85: 11/20, 0.90: 1/20, 0.95: 0/20

Table 18: Full supply PPO sweep after inclusion of all 120 completed refinement runs. Slopes are Firth estimates; no legacy Wald significance claims are retained.

λ	Runs	Midpoint and 95% CI	Slope	Usable
1	100	0.585 [0.544, 0.624]	−14.72	2000/2000
2	100	0.714 [0.683, 0.745]	−27.24	2000/2000
3	100	0.775 [0.755, 0.790]	−29.68	2000/2000
4	100	0.805 [0.780, 0.829]	−36.84	2000/2000
5	140	0.825 [0.812, 0.838]	−67.01	2000/2000
6	140	0.835 [0.816, 0.853]	−33.55	2000/2000
7	140	0.852 [0.840, 0.865]	−55.23	2000/2000

C Pre-registration and Deviations

E1–E4 were declared in a pre-registration written before any of their outcomes existed; E5, E6 and E7 were declared later, in response to reviews of an earlier draft, again before any of their runs. Each of E5–E7 had a pre-launch gate that reproduced a completed E1 run bit-for-bit through the new code path. Source, design and protocol hashes were frozen before E1 launched, and the E4 runner's hash before E4 launched, after a timing check. Launch scripts refused to run on any hash mismatch. The record shows four departures, none changing a design: (i) the frozen E1 runner omitted a protocol field read at evaluation, so every run crashed after training and before producing a label; the field was added with the mapping used by the earlier direct control, before any outcome existed; (ii–iii) two analysis-code errors (floating-point grid keys; the condition name "null" read as missing) raised exceptions before any fit was computed; (iv) found after the E4 analysis: the LBF label function reads its threshold from a calibration file that was absent from the frozen source and silently fell back to a placeholder of 0.5, so the stored E4 labels use $\theta = 0.5$ rather than the declared 0.626. Because every run stores its team return, all E4 results are recomputed at $\theta = 0.626$; the 0.5 analysis is kept as a sensitivity result (Table 26). The pre-declared predictions that failed are reported in the main text: the Adam-ϵ route (E2) two of three LBF predictions (E4), and in E7 the critic-target attribution (inconclusive), the moment control under the plain critic and the moment-matching check. An equality gate run before E1 reproduced the earlier platform's labels in 8/8 cells, but not its policies bit-for-bit, so archived runs were not reused.

D Refined Horizon Cohort

Table 19: E3 observed collapsed trajectories out of ten seeds per mixture cell at the four primary budgets. Full checkpoint records are in the archive.

Condition	Budget	w=0.2	w=0.4	w=0.6	w=0.65	w=0.7	w=0.75	w=0.8	w=0.85	w=1
Linear	300k	10	10	8	6	2	2	1	0	0
	900k	10	9	1	2	0	0	0	0	0
	1.8M	10	9	2	2	1	1	0	0	0
	3.6M	10	9	3	3	2	1	4	1	0
Curvature only	300k	10	10	6	1	0	0	0	1	0
	900k	10	7	2	1	0	0	0	0	0
	1.8M	10	8	3	3	0	0	0	0	0
	3.6M	10	10	3	3	0	0	1	0	1
$\lambda = 7$	300k	10	10	10	10	10	10	9	7	0
	900k	10	10	10	10	10	7	0	0	0
	1.8M	10	10	10	9	7	1	0	0	0
	3.6M	10	10	8	6	6	1	1	0	0

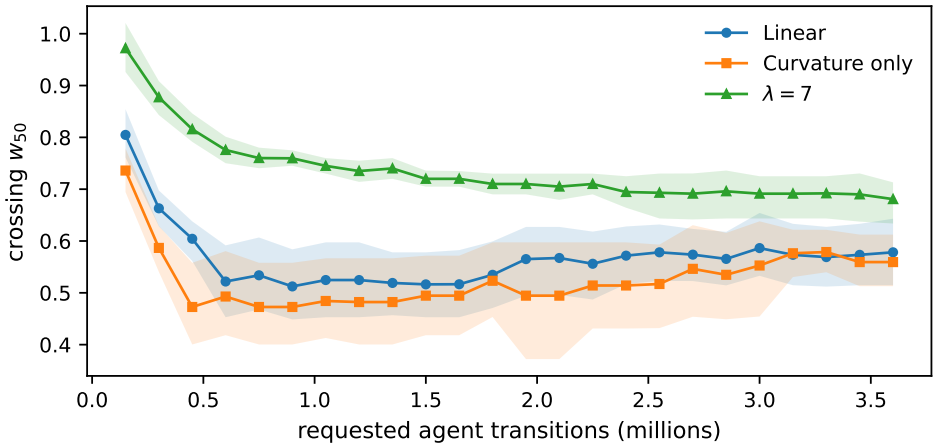


Figure 11: E3 crossings at every checkpoint (lines) with pointwise 95% seed-block bands. Default optimizer; nine mixture cells, ten seeds per cell and condition.

E Optimizer Factorial Details

Table 20: Change in each contrast relative to the default cell (global clip, raw reward), paired by seed. *: interval excludes zero.

Intervention	Δ(A−S)	Δ(A−B)	Δ(S−B)	Δ(R−B)
Global clip, normalized reward	+0.311* [+0.220, +0.402]	+0.003 [−0.130, +0.146]	−0.308* [−0.443, −0.158]	−0.195* [−0.349, −0.046]
Separate optimizers, raw	+0.248* [+0.136, +0.373]	+0.051 [−0.071, +0.161]	−0.197* [−0.366, −0.021]	−0.229* [−0.350, −0.104]
Separate optimizers, normalized	+0.311* [+0.209, +0.396]	−0.096 [−0.239, +0.034]	−0.407* [−0.565, −0.258]	−0.320* [−0.480, −0.174]
Global clip, raw, Adam ϵ=10^{-8}	+0.122* [+0.008, +0.253]	+0.023 [−0.087, +0.132]	−0.099 [−0.243, +0.033]	−0.090 [−0.201, +0.036]

Table 21: Optimizer diagnostics (means over runs of per-minibatch means). Pre-clip critic and actor gradient norms, post-clip (applied) actor norm, actor share of squared gradient norm, fraction of minibatches whose norm exceeds 0.5, and transformed-reward RMS. With separate optimizers no clipping is applied, so the applied actor norm equals the pre-clip norm.

Cell	Arm	Critic	Actor	Applied actor	Actor share	Over 0.5	Reward RMS
Global clip, raw (default)	B	1,326.81	2.07	0.0015	0.000	1.000	8.7
	A	3,425.24	1.83	0.0009	0.000	1.000	30.1
	S	11,964.76	1.46	0.0002	0.000	1.000	56.9
Global clip, normalized reward	B	0.12	4.39	0.4967	0.989	0.999	9.9
	A	0.14	4.22	0.4962	0.988	0.999	31.4
	S	0.12	4.25	0.4966	0.989	0.999	68.9
Separate optimizers, raw	B	1,285.95	4.06	4.0562	0.000	1.000	9.8
	A	3,737.64	3.63	3.6322	0.000	1.000	30.6
	S	13,923.98	4.41	4.4090	0.000	1.000	64.1
Separate optimizers, normalized	B	0.12	3.86	3.8594	0.986	0.999	9.8
	A	0.14	3.86	3.8567	0.985	0.999	31.5
	S	0.12	3.99	3.9904	0.987	0.999	69.5
Global clip, raw, Adam ϵ=10^{-8}	B	1,286.02	2.70	0.0019	0.000	1.000	9.0
	A	3,828.34	2.43	0.0011	0.000	1.000	31.1
	S	11,698.56	2.42	0.0003	0.000	1.000	59.5

Table 22: E1/E2 observed collapsed runs out of ten seeds per mixture cell at 300k.

Cell	Arm	w=0	w=0.2	w=0.4	w=0.5	w=0.6	w=0.7	w=0.85	w=1
Global clip, raw (default)	B	10	10	9	5	4	1	0	0
	A	10	10	10	10	10	10	3	0
	S	10	10	10	10	10	10	7	1
	R	10	10	10	10	10	10	1	0
Global clip, normalized reward	B	10	6	4	2	0	0	0	0
	A	10	10	9	8	4	3	0	0
	S	10	8	4	2	1	0	0	0
	R	10	10	3	1	0	0	0	0
Separate optimizers, raw	B	10	9	2	1	0	0	0	0
	A	10	10	10	10	9	2	0	0
	S	10	10	7	3	4	0	1	0
	R	10	9	3	2	0	0	0	0
Separate optimizers, normalized	B	10	10	5	2	1	0	1	0
	A	10	10	10	9	4	2	0	0
	S	10	8	6	1	0	0	0	0
	R	10	8	5	1	0	0	0	0
Global clip, raw, Adam ϵ=10^{-8}	B	10	10	5	4	1	2	0	0
	A	10	10	10	10	10	9	0	0
	S	10	10	10	9	9	4	2	0
	R	10	10	10	6	7	3	0	0

Table 23: Fraction of training rewards on the loss branch ($x - z_0 < 0$, $z_0 = -2.628$), by mixture weight, for the default and the fully decoupled E1 cells. The fraction depends on w and on the learned policy; near the crossings both branches are active.

Cell	Arm	w=0	w=0.2	w=0.4	w=0.5	w=0.6	w=0.7	w=0.85	w=1
Global clip, raw (default)	B	0.59	0.63	0.50	0.37	0.26	0.18	0.10	0.08
	A	0.59	0.65	0.58	0.57	0.55	0.49	0.13	0.07
	S	0.59	0.64	0.57	0.54	0.48	0.37	0.17	0.08
Separate optimizers, normalized	B	0.58	0.58	0.32	0.18	0.14	0.10	0.08	0.07
	A	0.58	0.62	0.51	0.38	0.24	0.17	0.09	0.08
	S	0.58	0.52	0.24	0.19	0.14	0.11	0.09	0.07

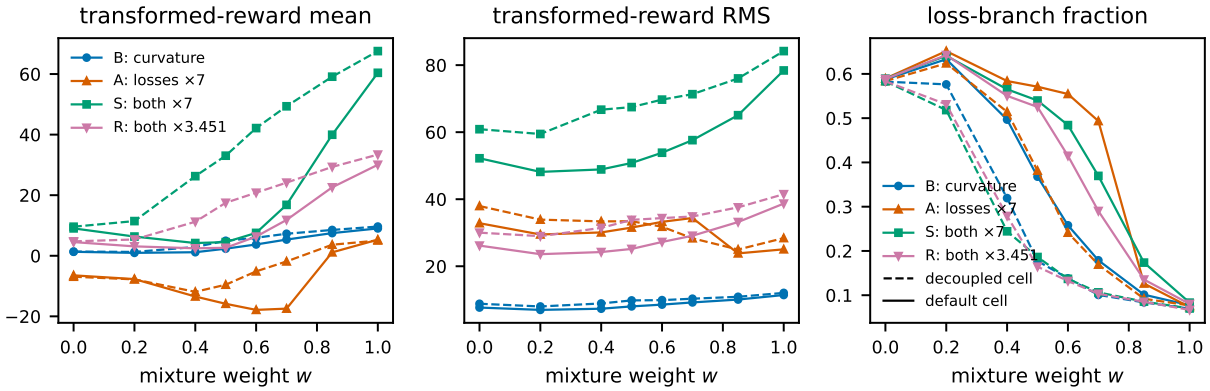


Figure 12: Transformed-reward moments over training in E1, by arm and mixture weight: mean, RMS and fraction on the loss branch (per-run means, averaged over seeds), for the default cell (solid) and the fully decoupled cell (dashed). The per-run tables keep these moments; per-rollout quantiles were also logged and are summarized for E7 in Table 24.

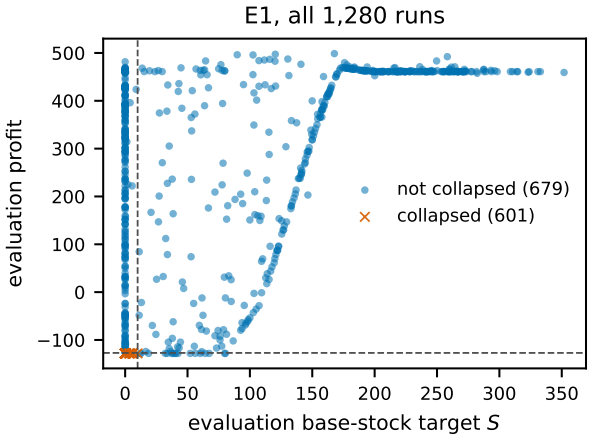


Figure 13: Collapse-label validation on all E1 runs: evaluation base-stock target $\bar{S}$ against evaluation profit, with the two label thresholds ($\bar{S} < 10$, profit ≤ -127.1) dashed. Collapsed runs sit at the lazy-policy corner.

Table 24: E7 training-reward and critic-target distributions. Per-rollout statistics of the transformed training reward (pooled over the five nodes) and of the rollout-buffer returns that the critic regresses, averaged over the 469 rollouts of each run and then over the 80 runs of each arm (ten seeds, eight cells). Quantile skewness is $(p_{90} + p_{10} - 2p_{50})/(p_{90} - p_{10})$. M's reward statistics are reconstructed exactly from the logged B reward and M's frozen affine map; its fraction of negative rewards is not recoverable from the logs. Descriptive; not pre-registered.

		Training reward (per rollout)							Critic targets	
Critic	Arm	Mean	SD	p_{10}	p_{50}	p_{90}	Q-skew	Frac. < 0	Mean	SD
Plain	B (curvature)	+5.5	7.0	−1.0	+5.8	+13.0	+0.01	0.23	+68	50
Plain	A (losses ×7)	−6.6	22.4	−25.7	−3.9	+12.2	−0.12	0.40	−104	113
Plain	S (both ×7)	+31.5	47.5	−15.1	+31.0	+88.9	+0.10	0.30	+256	270
Plain	M (affine of B)	−13.5	26.2	−38.9	−15.4	+20.0	+0.18	–	−185	148
PopArt	B (curvature)	+5.5	7.0	−1.2	+5.9	+13.0	−0.01	0.23	+73	53
PopArt	A (losses ×7)	−3.1	22.2	−21.0	+1.5	+12.6	−0.27	0.32	−70	123
PopArt	S (both ×7)	+38.5	48.6	−7.8	+40.4	+91.0	+0.01	0.24	+501	371
PopArt	M (affine of B)	−4.8	26.3	−29.2	−3.9	+22.4	−0.02	–	−90	146

F Level-Based Foraging Details

Table 25: E4 observed collapsed runs out of ten seeds per mixture cell (2M timesteps).

Optimizer	Arm	w=0	w=0.1	w=0.2	w=0.3	w=0.5	w=0.65	w=0.75	w=0.85	w=0.9	w=0.95
Global clip	Linear	4	1	1	3	0	0	1	0	0	0
	Curvature	9	9	3	2	2	2	0	0	0	0
	A (λ=4)	9	9	9	9	9	8	9	4	2	0
	S (×4)	9	8	7	4	5	1	0	0	0	0
Separate, no clip	Linear	7	6	0	2	1	0	0	0	0	0
	Curvature	6	9	7	2	2	1	0	0	0	0
	A (λ=4)	9	9	9	9	9	9	9	4	5	0
	S (×4)	9	8	8	5	6	0	0	0	0	0

Table 26: E4 sensitivity to the collapse threshold θ (native team return) and to the estimator. The stored run labels used $\theta = 0.5$ (deviation D4); the pre-declared $\theta = 0.626$ is primary. Last column: difference of decreasing-isotonic crossings.

θ	Optimizer	A	S	A−S (Firth)	A−S (isotonic)
0.500	Global clip	0.525 [0.188, 0.719]	0.107 [−0.065, 0.225]	+0.418 [+0.218, +0.553]	+0.575
	Separate, no clip	0.557 [0.250, 0.724]	0.077 [−0.126, 0.237]	+0.479 [+0.289, +0.613]	+0.731
0.626	Global clip	0.771 [0.584, 0.876]	0.324 [0.179, 0.451]	+0.447 [+0.312, +0.559]	+0.548
	Separate, no clip	0.843 [0.623, 0.918]	0.353 [0.224, 0.468]	+0.490 [+0.315, +0.619]	+0.325
0.750	Global clip	0.889 [0.812, 1.198]	0.519 [0.394, 0.610]	+0.370 [+0.259, +0.761]	+0.213
	Separate, no clip	0.921 [0.856, 1.097]	0.474 [0.364, 0.582]	+0.447 [+0.296, +0.694]	+0.354

Table 27: E4 fraction of training rewards below zero (reached through the LOAD effort cost), by mixture weight. Under A, agents largely stop attempting loads.

Optimizer	Arm	w=0	w=0.1	w=0.2	w=0.3	w=0.5	w=0.65	w=0.75	w=0.85	w=0.9	w=0.95
Global clip	A (λ=4)	0.012	0.014	0.013	0.014	0.018	0.020	0.021	0.042	0.059	0.112
	S (×4)	0.039	0.035	0.045	0.055	0.094	0.117	0.134	0.128	0.149	0.158
	Curvature	0.045	0.051	0.069	0.085	0.086	0.115	0.133	0.138	0.131	0.153
	Linear	0.089	0.100	0.105	0.141	0.165	0.176	0.169	0.166	0.165	0.176
Separate, no clip	A (λ=4)	0.012	0.012	0.012	0.012	0.013	0.015	0.017	0.040	0.046	0.100
	S (×4)	0.034	0.039	0.042	0.054	0.070	0.120	0.130	0.123	0.133	0.160
	Curvature	0.042	0.042	0.054	0.075	0.096	0.108	0.122	0.132	0.143	0.143
	Linear	0.066	0.076	0.118	0.137	0.161	0.172	0.169	0.162	0.157	0.175

G Computational Accounting

E1–E7 comprise 4,060 trajectories and about 1,422 single-core hours (Table 28), excluding smoke tests, equality gates and extraction. The legacy sweeps and the earlier confirmation cohorts were run separately on another platform and are not included.

Table 28: Measured v4 compute: completed Slurm array tasks, one CPU core each (single-thread PyTorch), summed elapsed time. Runs were executed at most eight concurrently.

ID	Experiment	Trajectories	Core-hours	Mean minutes per run
E1	Optimizer × normalization factorial, 300k	1,280	177.5	8.3
E2	Adam-ϵ addendum, 300k	320	44.4	8.3
E3	Refined horizon cohort to 3.6M	270	449.4	99.9
E4	MAPPO on LBF, 2M timesteps	800	143.2	10.7
E5	Decoupled horizon cohort to 3.6M	270	450.4	100.1
E6	Clipping isolation, 300k	720	101.4	8.4
E7	Critic-target normalization and moment control, 300k	400	56.0	8.4
	Total	4,060	1,422.4	

H Reproducibility Artifact

The anonymized archive, provided as supplementary material with this submission, contains the manuscript source; the frozen designs, pre-registration, hashes and deviation log; the executed source for E1–E7; per-run tables for every E1–E7 trajectory (protocol, label, evaluation summary and optimizer diagnostics); the extraction and analysis scripts; and the renderer that produces every v4 table, figure and number. Labels are never taken from a stored field without check: one module (`analysis/labels.py`) recomputes them from stored evaluation statistics, asserts each declared threshold against the sentence of the pre-registration that declares it, rejects undeclared thresholds and out-of-range inputs, and verifies that every stored supply-chain label equals the declared rule; unit tests cover these checks and the E4 relabelling counts. The archive also contains the pre-registration and runner for the outstanding experiment E8 (Section 10), which has not been run. Retraining every trajectory is not part of the one-command rebuild: it regenerates all analyses, tables, figures and the PDF from the per-run tables, and the executed source and frozen designs allow any run to be repeated. Raw per-minibatch diagnostics and checkpoints (about 45 GB) are available on request. The legacy sweeps are accompanied by their per-run records, without a preserved execution environment.